Methods Paper - Chapter 1

Single-cell RNA-seq preprocessing, clustering, bibliography annotation, and subcluster marker inspection

SingleCell Pipeline

2026-07-15

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This methods-paper R Markdown documents the current Chapter 1 workflow only. It is intentionally safe to knit: code blocks are displayed but not executed. To run the analysis, execute the corresponding sections from `workflow/capitulo1_single_cell.Rmd` inside the Docker container.

The active annotation path is bibliography-based (`biblio_marks_custom.txt` -> `pbmc_harmony$celltype`). Reference-transfer annotation is not part of this draft. The document stops after the optional subcluster marker-inspection section; h5ad export, pseudobulk, DESeq2, GO, network, and pseudotime analyses are intentionally excluded for now.

0.1 Materials and data preparation

The workflow is designed to run inside the Docker image `matigara/scrnaseq:latest`, with the project directory mounted as `/workspace`. In the current server setup, the relevant host paths are:

```
/projects/aivvy/SingleCell/workflow/capitulo1_single_cell.Rmd
/projects/aivvy/SingleCell/workflow/ScRNA_Analysis_Functions.R
/projects/aivvy/resultados/
```

0.1.1 Minimal components required to run the pipeline

The minimum inputs for Chapter 1 are:

1. A working Docker installation.
2. The SingleCell workflow files and helper functions.
3. One Cell Ranger `filtered_feature_bc_matrix/` directory per sample.
4. A bibliography marker table with columns `cell types` and `gene`.
5. Enough local or server memory to hold the merged Seurat object.

For this run, the marker table used by the workflow is:

```
/workspace/biblio_marks_custom.txt
```

The marker-table order is biologically meaningful. The dotplot helper preserves marker-table row order on the Y axis, while the X axis is ordered according to the cell-type order implied by that same marker table. Labels with numeric suffixes, such as `Epidermis Hypocotyl.1` and `Epidermis Hypocotyl.2`, are kept adjacent to their parent marker category instead of being moved to the end.

0.1.2 Software and package versions

The table below records the versions available in the active Docker container used for this methods draft.

Component	Version
R	4.5.3
Python	3.12.3
Seurat	5.5.0
SeuratDisk	0.0.0.9021
harmony	2.0.3
DoubletFinder	2.0.6
clustree	0.5.1
ggplot2	4.0.3
dplyr	1.2.1
patchwork	1.3.2
zellkonverter	1.20.1
scanpy	1.12.1
anndata	0.12.16
numpy	2.4.6
pandas	2.3.3

0.1.3 Docker startup

For a new local run, start the container from the host folder that contains the data, workflow, and output directory:

```
docker run -it --rm \
  -v ~/your-workdir:/workspace \
  matigara/scrnaseq:latest /bin/bash
```

On the current server, the Docker container is already running and the project is mounted at `/workspace`. From inside the container, start R and run the workflow by sections:

```
R
```

0.1.4 Raw data download and Cell Ranger processing

This Chapter 1 workflow starts from Cell Ranger output, specifically the `outs/filtered_feature_bc_matrix/` folder for each sample. If only raw FASTQ files are available, first download or stage the FASTQs, then run Cell Ranger for each sample before starting the R workflow.

In the archived Chapter 0 run available on the server, raw FASTQ files were not present at execution time. The script therefore validated existing Cell Ranger outputs and copied the Cell Ranger metrics summaries. The validated run used this layout:

```
/workspace/cellranger/Sample_0N/outs/filtered_feature_bc_matrix/
/workspace/cellranger/Sample_05N/outs/filtered_feature_bc_matrix/
/workspace/cellranger/Sample_5N/outs/filtered_feature_bc_matrix/
```

The validation script used for that run is kept in the repository as:

```
workflow/runs/cap0_r1.sh
```

Its logic checks, for each sample, that the following Cell Ranger files exist and are non-empty:

```
outs/filtered_feature_bc_matrix/matrix.mtx.gz
outs/filtered_feature_bc_matrix/barcodes.tsv.gz
outs/filtered_feature_bc_matrix/features.tsv.gz
outs/metrics_summary.csv
```

For a new analysis starting from public FASTQ files, first create a FASTQ staging directory and download the reads. The exact accession list depends on the experiment, but the workflow expects one folder per sample after download. Common download patterns are:

```
# Option A: ENA/URL download when direct FASTQ links are available
mkdir -p /workspace/fastq/ScWT
wget -c -P /workspace/fastq/ScWT URL_TO_R1_FASTQ.gz
wget -c -P /workspace/fastq/ScWT URL_TO_R2_FASTQ.gz

# Option B: SRA Toolkit when SRR accessions are provided
prefetch SRR_ACCESSION
fasterq-dump SRR_ACCESSION --split-files --threads 8 --outdir /workspace/fastq/ScWT
gzip /workspace/fastq/ScWT/*.fastq
```

After FASTQs are available, run Cell Ranger for each sample. Cell Ranger is not bundled inside the Docker image used for the downstream R/Python workflow, so it should be run on the host, a server, or an HPC environment where Cell Ranger and the appropriate reference transcriptome are installed.

Generic Cell Ranger command structure:

```
cellranger count \
  --id ScWT \
  --transcriptome /path/to/reference_transcriptome \
  --fastqs /path/to/fastq_directory \
  --sample ScWT \
  --localcores 16 \
  --localmem 128
```

Use the resulting `filtered_feature_bc_matrix/` path in the sample manifest in Section 1.

0.1.5 Cluster or server requirement

A cluster is not strictly required for small pilot datasets, but server or HPC execution is recommended once the number of cells, samples, or parameter sweeps increases. The most memory-intensive steps are merging samples, scaling, PCA/UMAP, neighbor construction, marker detection, and subclustering. Cell Ranger also benefits from multiple CPU cores and high memory, especially when processing raw FASTQ files.

0.2 Configuration

Edit this block before running the pipeline.

```
# CONFIGURATION - Edit only this block before running the pipeline
# ============================================================================

# Directory containing the pipeline helper scripts.
# Inside the Docker container this is typically /workspace/SingleCell
PIPELINE_DIR <- "/workspace/SingleCell/workflow"

# Root directory for your project data and results.
# All result files will be written to DATA_DIR/resultados/<step>/
DATA_DIR   <- "/workspace/."
base_dir   <- file.path(DATA_DIR, "resultados")

# ============================================================================
```

0.3 Initialization

Load helper scripts, set reproducibility options, and create output directories.

```
# INITIALIZATION
# ============================================================================

# -- Load helper scripts ---------------------------------------------------
# Each file is fully documented at https://github.com/clifford2001/SingleCell
source(file.path(PIPELINE_DIR, "load_libraries.R"))          # all R packages
source(file.path(PIPELINE_DIR, "ScRNA_Analysis_Functions.R"))# analysis functions

set.seed(1807)
options(Seurat.allow.s4 = FALSE)  # required for Seurat 5 compatibility
setwd(DATA_DIR)

# -- Create per-step output directories ------------------------------------
list2env(create_pipeline_dirs(base_dir), envir = .GlobalEnv)  # creates output folders and loads their

# output_dir is the global variable used by save_pdf / save_qc / save_vln helpers.
# It is reassigned at the start of each section to the appropriate step directory.
output_dir <- base_dir


# ============================================================================
# ########################  PART 1 - SINGLE-CELL ANALYSIS  ###################
# ============================================================================
```

0.4 Section 1 - DATA LOADING AND PRE-FILTER QC

```
# SECTION 1 - DATA LOADING AND PRE-FILTER QC
# ============================================================================
# Each sample is loaded from its input file and mitochondrial / chloroplast
# read fractions are computed per cell to guide filtering thresholds in
# Section 2. Pre-filter violin plots are saved to 01_qc/.
#
# +- CHANGE FOR YOUR ORGANISM ------------------------------------------------
#   Arabidopsis : mt_pattern = "^ATMG"  |  cp_pattern = "^ATCG"
#   Human       : mt_pattern = "^MT-"   |  cp_pattern = NULL
#   Mouse       : mt_pattern = "^mt-"   |  cp_pattern = NULL
# +---------------------------------------------------------------------------
output_dir <- dir_01

# -- Sample manifest (CellRanger filtered_feature_bc_matrix) ------------------
# Add one entry per sample. Each entry needs:
#   file      - path to the filtered_feature_bc_matrix/ directory (relative to DATA_DIR)
#   label     - unique name for this sample (appears in all plots)
#   condition - experimental group this sample belongs to
samples <- list(
  list(file = "cellranger_v2/ScWT/outs/filtered_feature_bc_matrix",  label = "WT",  condition = "WT"),
  list(file = "cellranger_v2/Scpifq/outs/filtered_feature_bc_matrix", label = "pifq", condition = "pifq


# -- Plot colors (one color per sample label) ---------------------------------
colors <- c(
  "WT"  = "#66c2a5",
  "pifq"  = "#fc8d62"
)


mt_pattern <- "^ATMG"  # Arabidopsis mitochondrial genes
cp_pattern <- "^ATCG"  # Arabidopsis chloroplast genes


# Load all samples using the helper function
seurat_list_raw <- load_seurat_samples(samples = samples,
                                       DATA_DIR = DATA_DIR,
                                       mt_pattern = mt_pattern,
                                       cp_pattern = cp_pattern)

plot_qc_batch(seurat_list_raw, colors, "qc_prefilter.pdf")


# ============================================================================

message("\nOK SECTION 1 COMPLETE: QC pre-filter plots saved")
```

Representative output from Section 1:

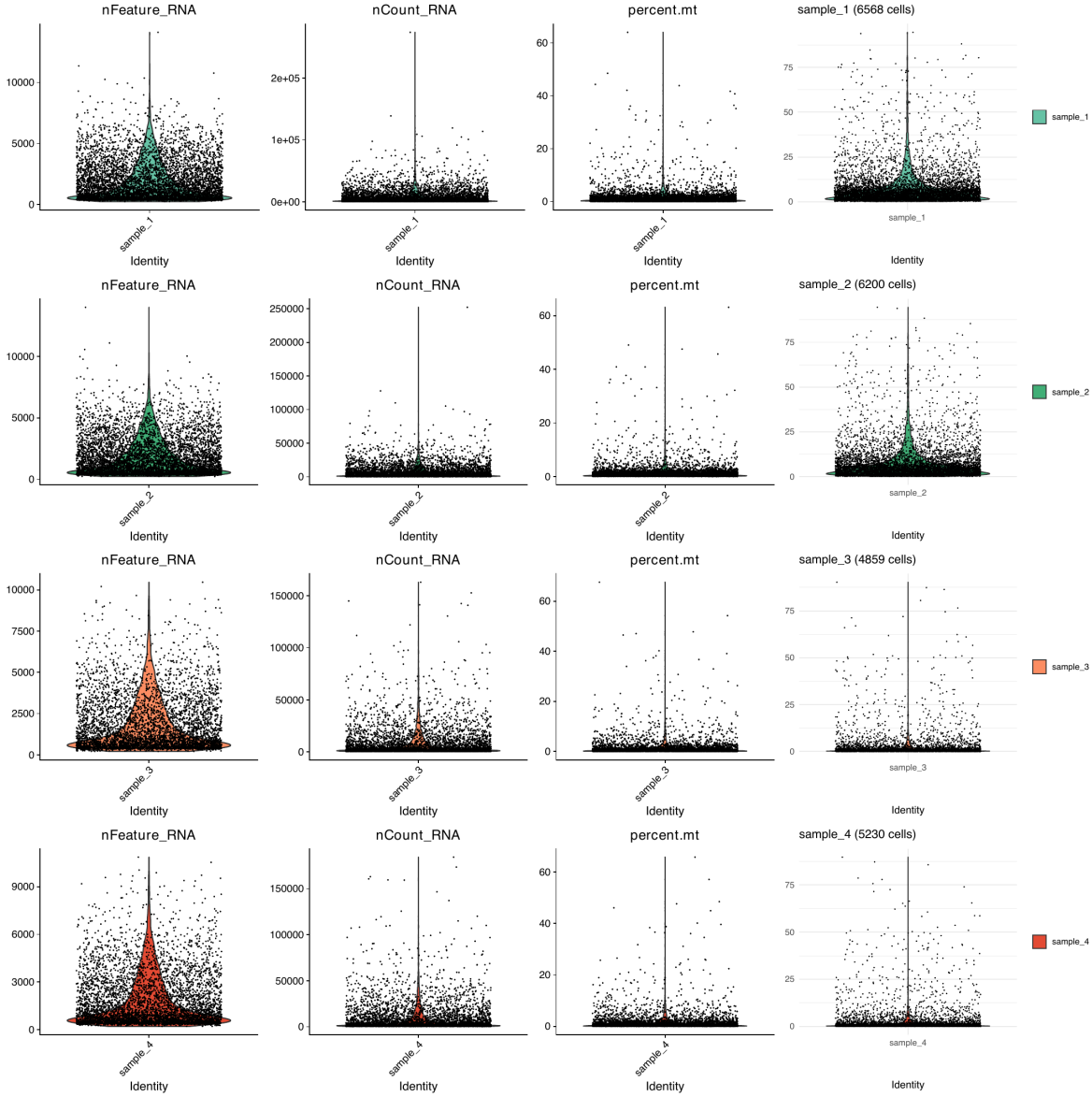


Figure 1: Pre-filter QC violin plots.

0.5 Section 2 - CELL FILTERING AND DOUBLET DETECTION

```
# SECTION 2 - CELL FILTERING AND DOUBLET DETECTION
# ============================================================================
# Thresholds are set based on the pre-filter violin plots (01_qc/qc_prefilter.pdf).
# Cells below min_features or above max_mt are removed. DoubletFinder is
# applied per sample to discard putative doublets.
#
# +- ADJUST THRESHOLDS AFTER INSPECTING 01_qc/qc_prefilter.pdf -----------------
#   min_features : minimum number of detected genes per cell (default 200)
#   max_mt       : maximum mitochondrial read percentage  (default 5 %)
# +----------------------------------------------------------------------------
output_dir <- dir_01   # both pre- and post-filter QC plots go to 01_qc/

seurat_list <- filter_seurat_samples(seurat_list_raw, min_features = 200, max_mt = 5)

plot_qc_batch(seurat_list, colors, "qc_postfilter.pdf")

# Checkpoint - restore with: seurat_list <- readRDS("resultados/objects/seurat_list_postfilter.rds")
saveRDS(seurat_list, file.path(dir_objects, "seurat_list_postfilter.rds"))


# ============================================================================

message("\nOK SECTION 2 COMPLETE: Filtering and doublet detection complete")
```

Representative output from Section 2:

0.6 Section 3 - MERGE AND INITIAL PREPROCESSING

```
# SECTION 3 - MERGE AND INITIAL PREPROCESSING
# ============================================================================
# Filtered samples are merged and preprocessed: log-normalization, variable
# feature selection (VST, 2,000 features), scaling, PCA (30 PCs), and UMAP.
# The resulting UMAP shows batch effects before integration.

output_dir <- dir_01

pbmc_harmony <- reduce(seurat_list, merge) %>%  # merge all samples into one object
  NormalizeData(verbose = FALSE) %>%
  FindVariableFeatures(selection.method = "vst", nfeatures = 2000, verbose = FALSE) %>%
  ScaleData(verbose = FALSE) %>%
  RunPCA(npcs = 30, verbose = FALSE) %>%
  RunUMAP(reduction = "pca", dims = 1:30, verbose = FALSE)

save_pdf(DimPlot(pbmc_harmony, group.by = "orig.ident", cols = colors),
         "umap_preharmony.pdf")

# Checkpoint - restore with: pbmc_harmony <- readRDS("resultados/objects/pbmc_harmony_preharmony.rds")
saveRDS(pbmc_harmony, file.path(dir_objects, "pbmc_harmony_preharmony.rds"))


# ============================================================================
```

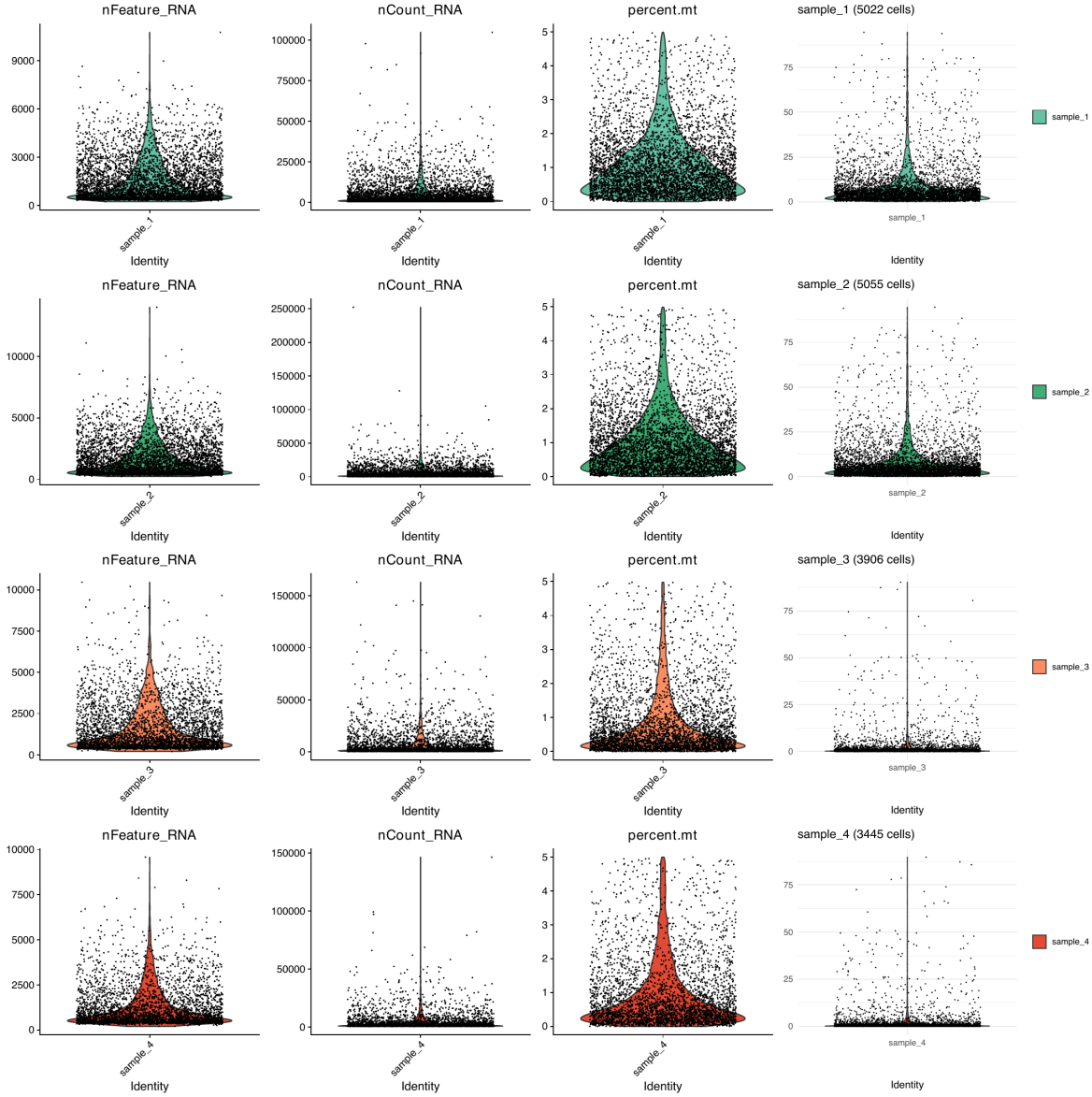


Figure 2: Post-filter QC violin plots.

```r
message("\nOK SECTION 3 COMPLETE: Merge and preprocessing complete")
```

Representative output from Section 3:

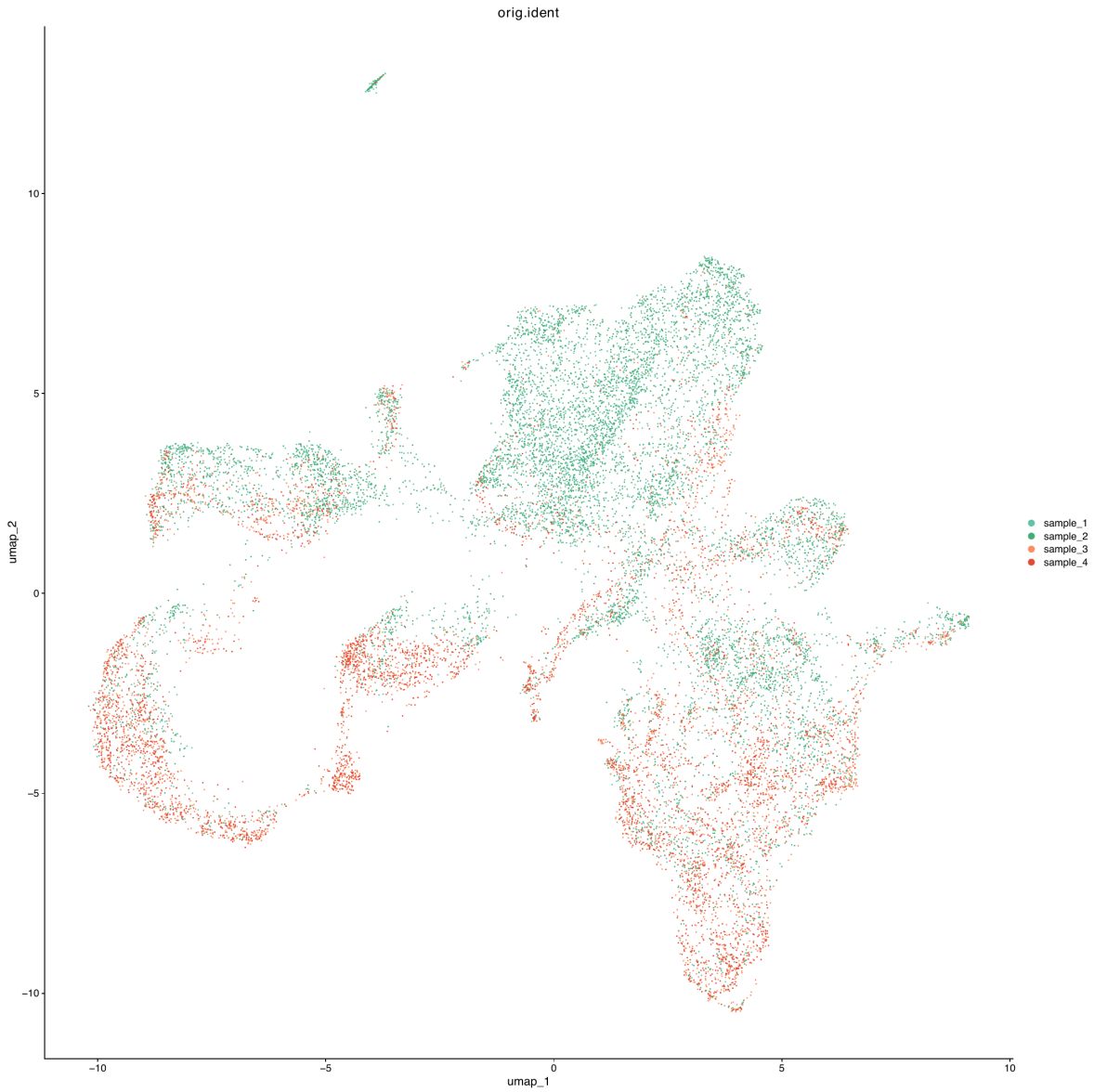


Figure 3: UMAP before Harmony correction, colored by sample.

0.7 Section 4 - HARMONY BATCH CORRECTION

```r
# SECTION 4 - HARMONY BATCH CORRECTION
# ==============================================================================
# Harmony adjusts cell embeddings to remove sample-level batch effects while
# preserving biological variation. All downstream steps use the "harmony"
# reduction instead of "pca".

output_dir <- dir_01
```

```r
pbmc_harmony <- pbmc_harmony %>%
  RunHarmony("orig.ident", plot_convergence = FALSE) %>%
  RunUMAP(reduction = "harmony", dims = 1:30, verbose = FALSE)

save_pdf(DimPlot(pbmc_harmony, group.by = "orig.ident", cols = colors),
         "umap_postharmony.pdf")

# Checkpoint - restore with: pbmc_harmony <- readRDS("resultados/objects/pbmc_harmony_postharmony.rds")
saveRDS(pbmc_harmony, file.path(dir_objects, "pbmc_harmony_postharmony.rds"))


# ============================================================================

message("\nOK SECTION 4 COMPLETE: Harmony integration complete")
```

Representative output from Section 4:

0.8 Section 5 - RESOLUTION OPTIMIZATION

```r
# SECTION 5 - RESOLUTION OPTIMIZATION
# ============================================================================
# Clustree tracks how Leiden communities split or merge across candidate
# resolutions. Choose the lowest resolution where clusters stabilize.
#
# +- PARAMETERS ------------------------------------------------------------
#   resolutions_test : Leiden resolutions swept by clustree
#   -> Inspect elbow_plot.pdf and clustree2.pdf before setting cluster_resolution in Section 6.
# +-------------------------------------------------------------------------
resolutions_test <- c(0.15, 0.35, 0.45, 0.55, 1.0)

output_dir <- dir_02

# Elbow plot: k-means within-cluster sum of squares across candidate k values.
# This is only a diagnostic plot; it does not change the clustering parameters.
k_range <- 1:31
pca_data <- Embeddings(pbmc_harmony, "pca")[, 1:30]
wss <- sapply(
  k_range,
  function(k) kmeans(pca_data, centers = k, nstart = 4)$tot.withinss
)

elbow_plot <- ggplot(data.frame(k = k_range, wss = wss), aes(k, wss)) +
  geom_line() +
  geom_point() +
  labs(
    x = "Number of clusters (k)",
    y = "Within-cluster sum of squares"
  ) +
  theme_minimal()

save_pdf(elbow_plot, "elbow_plot.pdf", w = 18, h = 18)

clu <- pbmc_harmony %>%
```

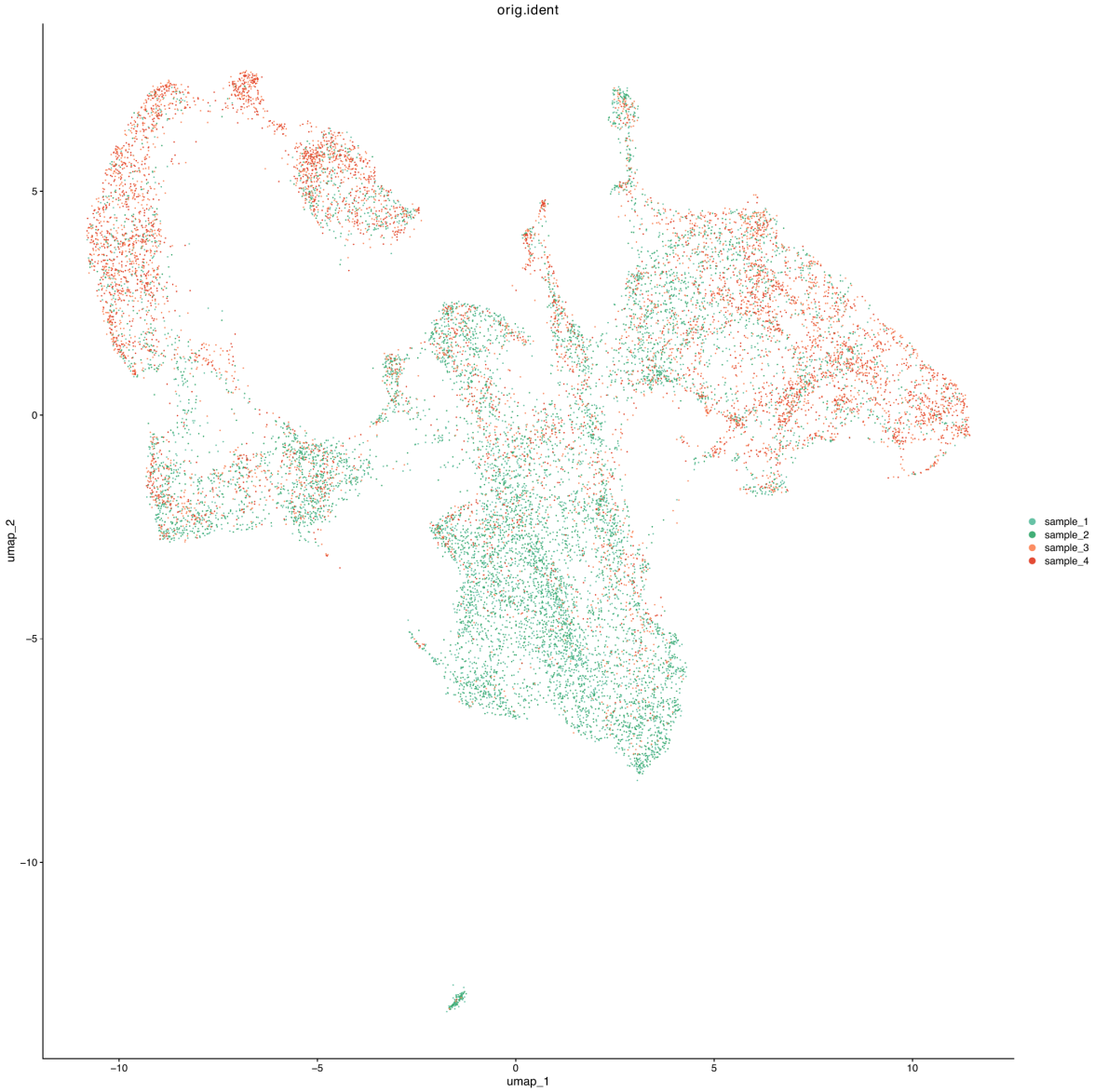


Figure 4: UMAP after Harmony correction, colored by sample.

```
  RunUMAP(reduction = "harmony", dims = 1:30, verbose = FALSE) %>%
  FindNeighbors(reduction = "harmony", dims = 1:30,
                k.param = 20, verbose = FALSE)

for (res in resolutions_test)
  clu <- FindClusters(clu, resolution = res, algorithm = 4, verbose = FALSE)

save_pdf(clustree(clu, prefix = "RNA_snn_res."), "clustree2.pdf", w = 18, h = 18)


# ==============================================================================

message("\nOK SECTION 5 COMPLETE: Elbow plot and clustree saved - inspect elbow_plot.pdf and clustree2.p
```

Representative outputs from Section 5:

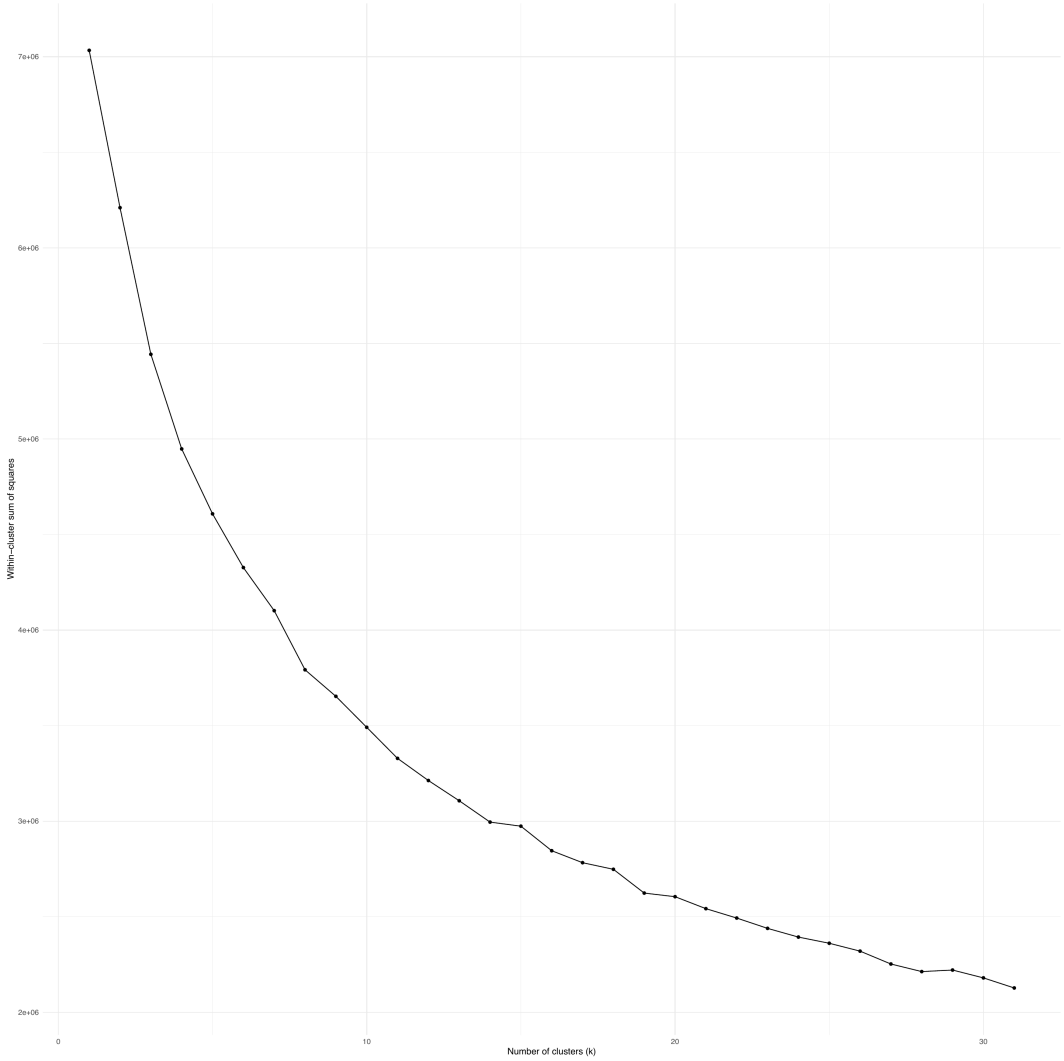


Figure 5: Elbow diagnostic used to inspect candidate cluster numbers.

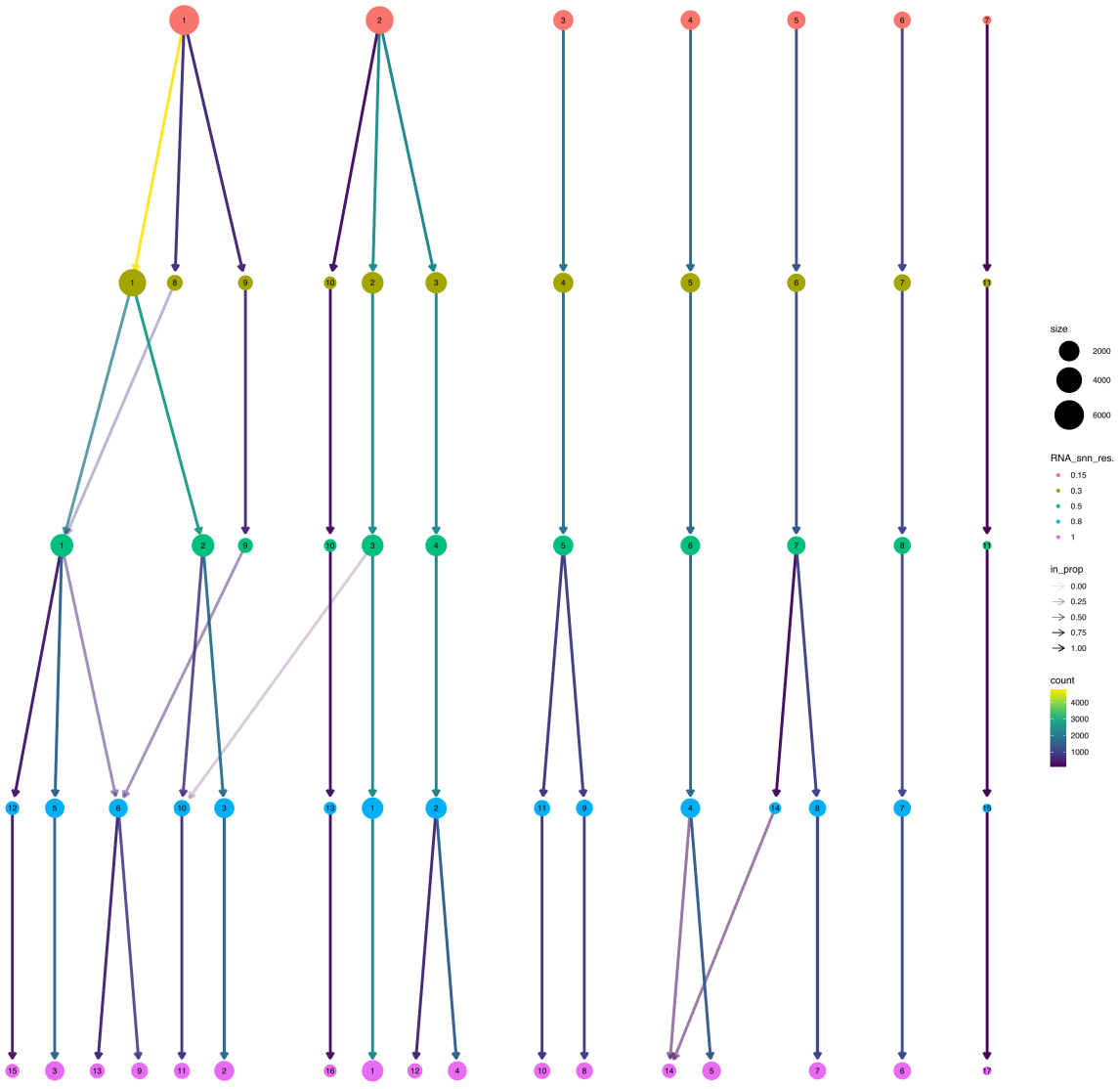


Figure 6: Cluster tree across candidate Leiden resolutions.

0.9 Section 6 - FINAL CLUSTERING

```
# SECTION 6 - FINAL CLUSTERING
# ============================================================================
# Apply the selected resolution for the final cluster assignment.
# After clustering, a UMAP coloured by Seurat cluster and a bar chart of
# cells per sample are saved to 02_clustering/.
#
# +- SET RESOLUTION AFTER INSPECTING clustree.pdf ----------------------------
#   cluster_resolution : Leiden resolution for final clustering (default 0.3)
# +--------------------------------------------------------------------------
cluster_resolution <- 0.35
output_dir <- dir_02

# clu (Section 5) was temporary - re-run on pbmc_harmony to embed the final clusters
pbmc_harmony <- pbmc_harmony %>%
  RunUMAP(reduction = "harmony", dims = 1:30, verbose = FALSE) %>%
  FindNeighbors(reduction = "harmony", dims = 1:30,
                k.param = 20, verbose = FALSE) %>%
  FindClusters(resolution = cluster_resolution, algorithm = 4, verbose = FALSE)

Idents(pbmc_harmony) <- "seurat_clusters"
save_pdf(DimPlot(pbmc_harmony, group.by = "seurat_clusters", label = TRUE),
         "umap_seuratclusters.pdf")


# ============================================================================

message("\nOK SECTION 6 COMPLETE: Final clustering complete")
```

Representative output from Section 6:

0.10 Section 7 - CELL-TYPE ANNOTATION

```
# SECTION 7 - CELL-TYPE ANNOTATION
# ============================================================================
# Cell-type labels are assigned via bibliography-based markers
# (biblio_marks.txt): the marker table read above is crossed with
# cluster-level differential genes to assign initial labels automatically.
# Result stored in pbmc_harmony$celltype.
#
# A second strategy - reference transfer from a published Seurat object
# (e.g. an Arabidopsis leaf atlas) via FindTransferAnchors/TransferData - is
# not run by default. The full block, plus the downstream sections wired to
# it, is preserved in capitulo1_single_cell_reference_backup.R.

output_dir <- dir_03

biblio_marks_file <- file.path(DATA_DIR, "biblio_marks_custom.txt")
marker_table      <- read.table(biblio_marks_file, header = TRUE, sep = "\t", quote = "")

# -- Bibliography-based annotation ---------------------------------------------
markers <- find_markers(pbmc_harmony,
                        output_file = file.path(output_dir, "FindAllMarkers.tsv"))
```

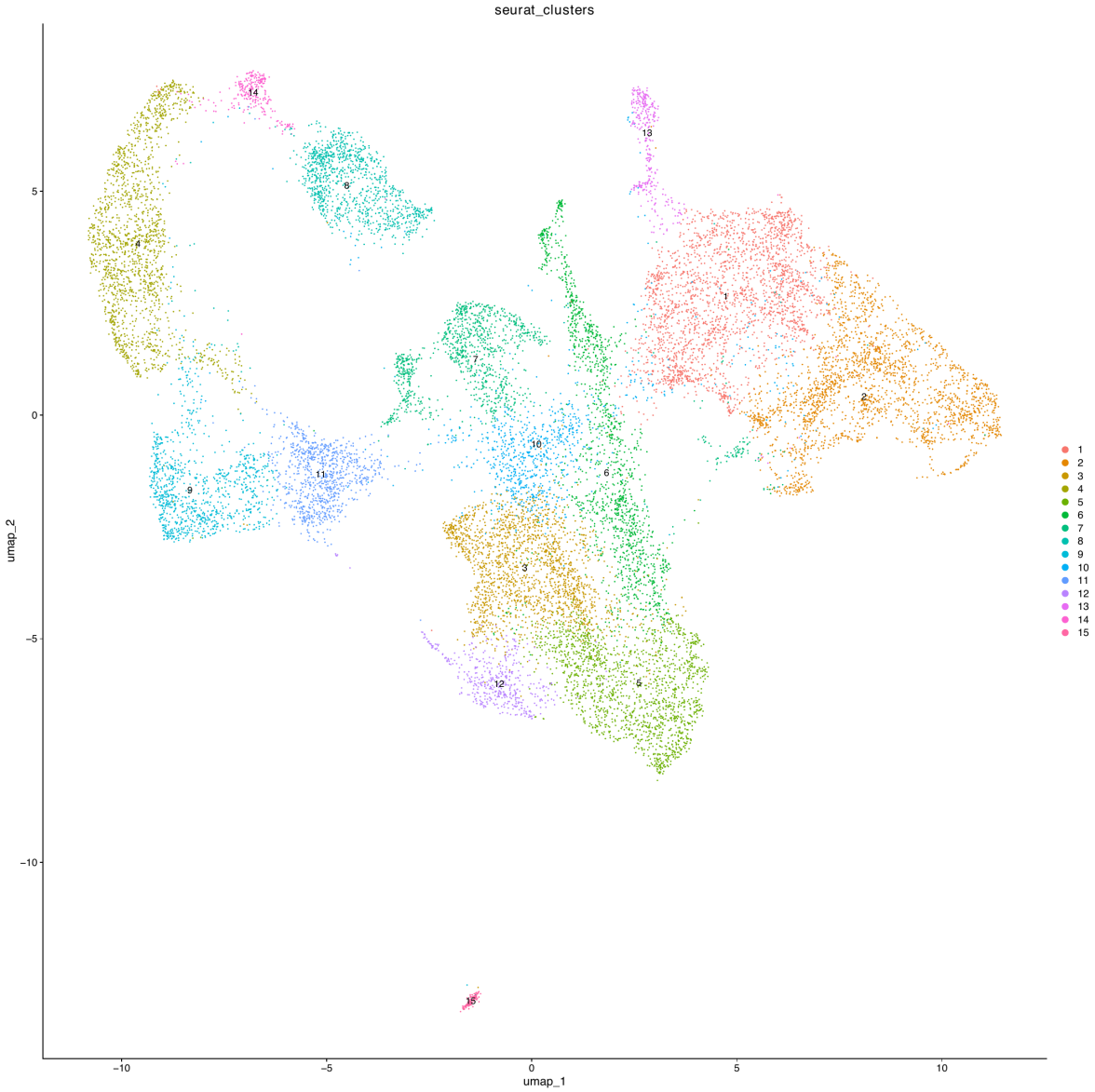


Figure 7: Final UMAP colored by numbered Seurat clusters.

```r
pbmc_harmony <- annotate_by_markers(pbmc_harmony, markers,
                                    reference_file = biblio_marks_file)
# Annotation stored in: pbmc_harmony$celltype

plot_marker_dotplot(
  pbmc_harmony,
  marker_table,
  annot_col = "celltype", # uses the newly assigned annotation column
  outfile   = file.path(output_dir, "dotplot_marker_table_annotation_biblio.pdf"),
  width = 18, height = 18
)

save_pdf(DimPlot(pbmc_harmony, group.by = "celltype",
                 label = TRUE, repel = TRUE, raster = FALSE),
         "umap_annotation_biblio.pdf")


# ==============================================================================

message("\nOK SECTION 7 COMPLETE: Cell-type annotation complete")
```

Representative outputs from Section 7:

0.11 Section 8 - ANNOTATED CLUSTREE

```r
# SECTION 8 - ANNOTATED CLUSTREE
# ==============================================================================
# Reuses the resolution sweep object from Section 5 and overlays the final
# bibliography-based annotation on each node.

output_dir <- dir_03

# Mode: returns the most frequent annotation in each clustree node.
Mode <- function(x) {
  x <- as.character(x)
  x <- x[!is.na(x) & nzchar(x)]
  if (length(x) == 0) return(NA_character_)
  names(sort(table(x), decreasing = TRUE))[1]
}

stopifnot(exists("clu"))
stopifnot("celltype" %in% colnames(pbmc_harmony@meta.data))

celltype_label <- as.character(pbmc_harmony$celltype)
names(celltype_label) <- Cells(pbmc_harmony)
clu$celltype_label <- celltype_label[Cells(clu)]

print(table(clu$celltype_label, useNA = "ifany"))

save_pdf(
  clustree(clu, prefix = "RNA_snn_res.",
           node_label = "celltype_label", node_label_aggr = "Mode"),
  "clustree_annotated.pdf", w = 14, h = 14
```

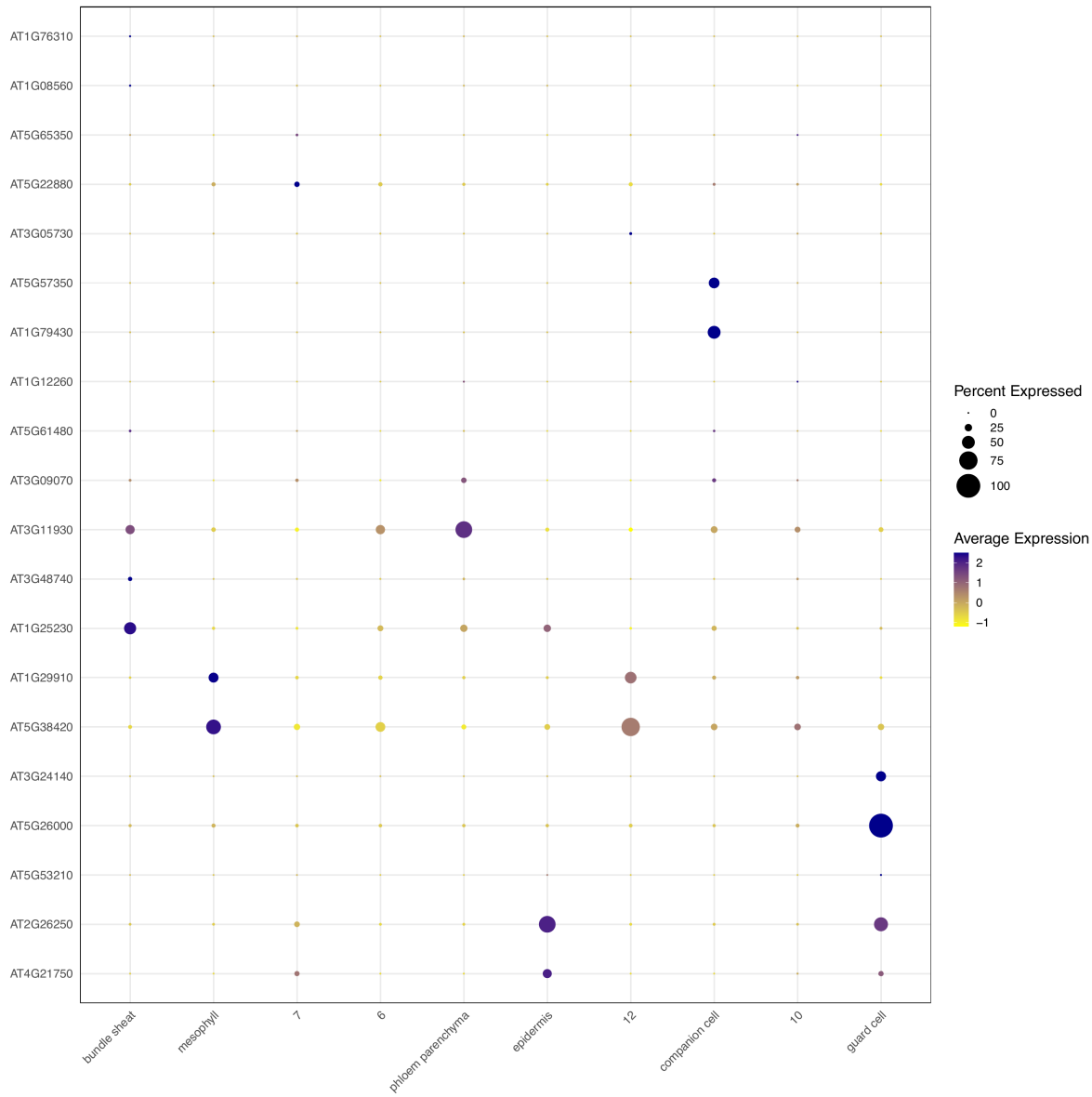


Figure 8: Marker dotplot after bibliography-based cell-type annotation.

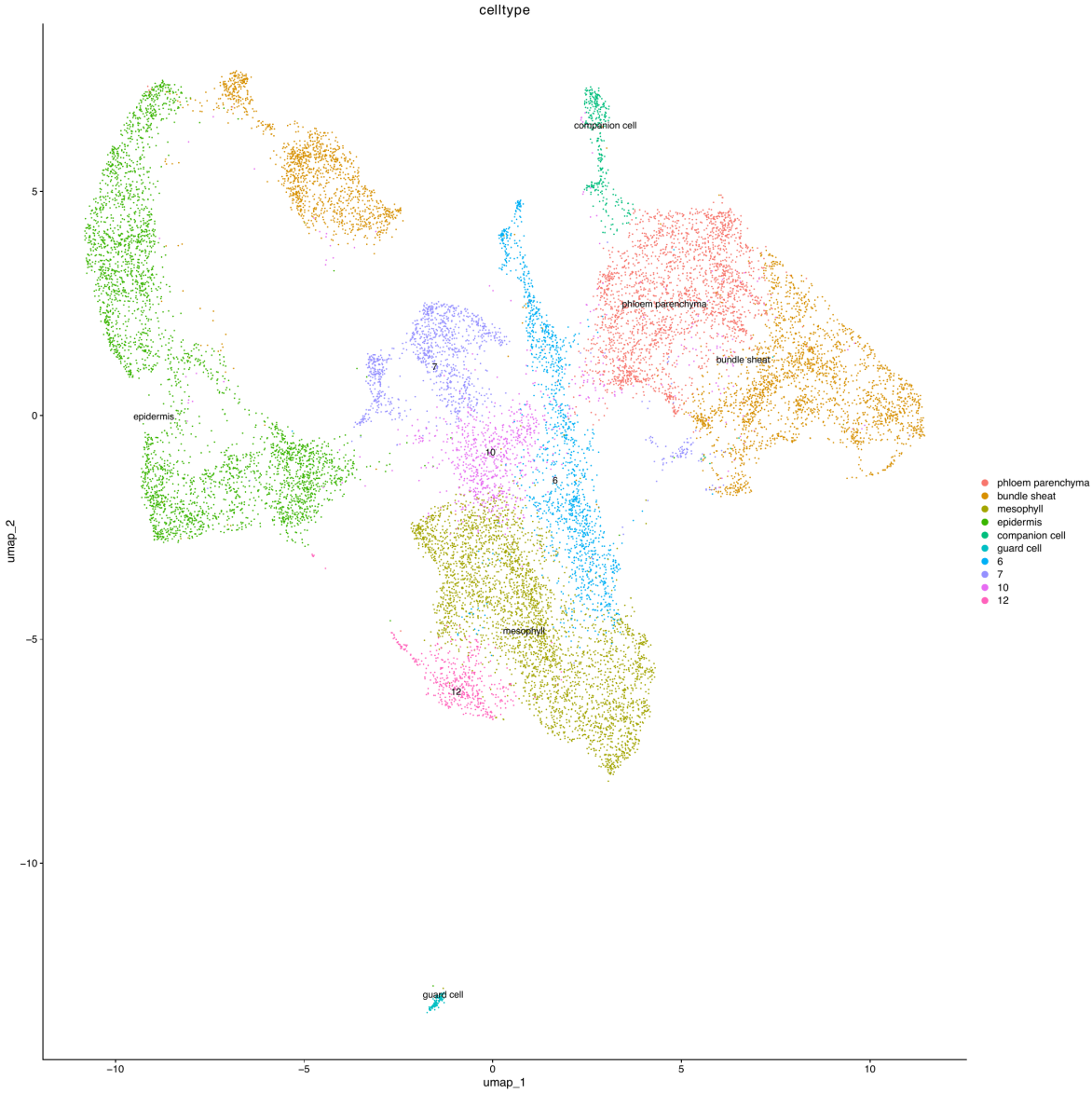


Figure 9: UMAP colored by bibliography-based cell-type annotation.

```
)


# ============================================================================

message("\nOK SECTION 8 COMPLETE: Annotated clustree saved")
```

Representative output from Section 8:

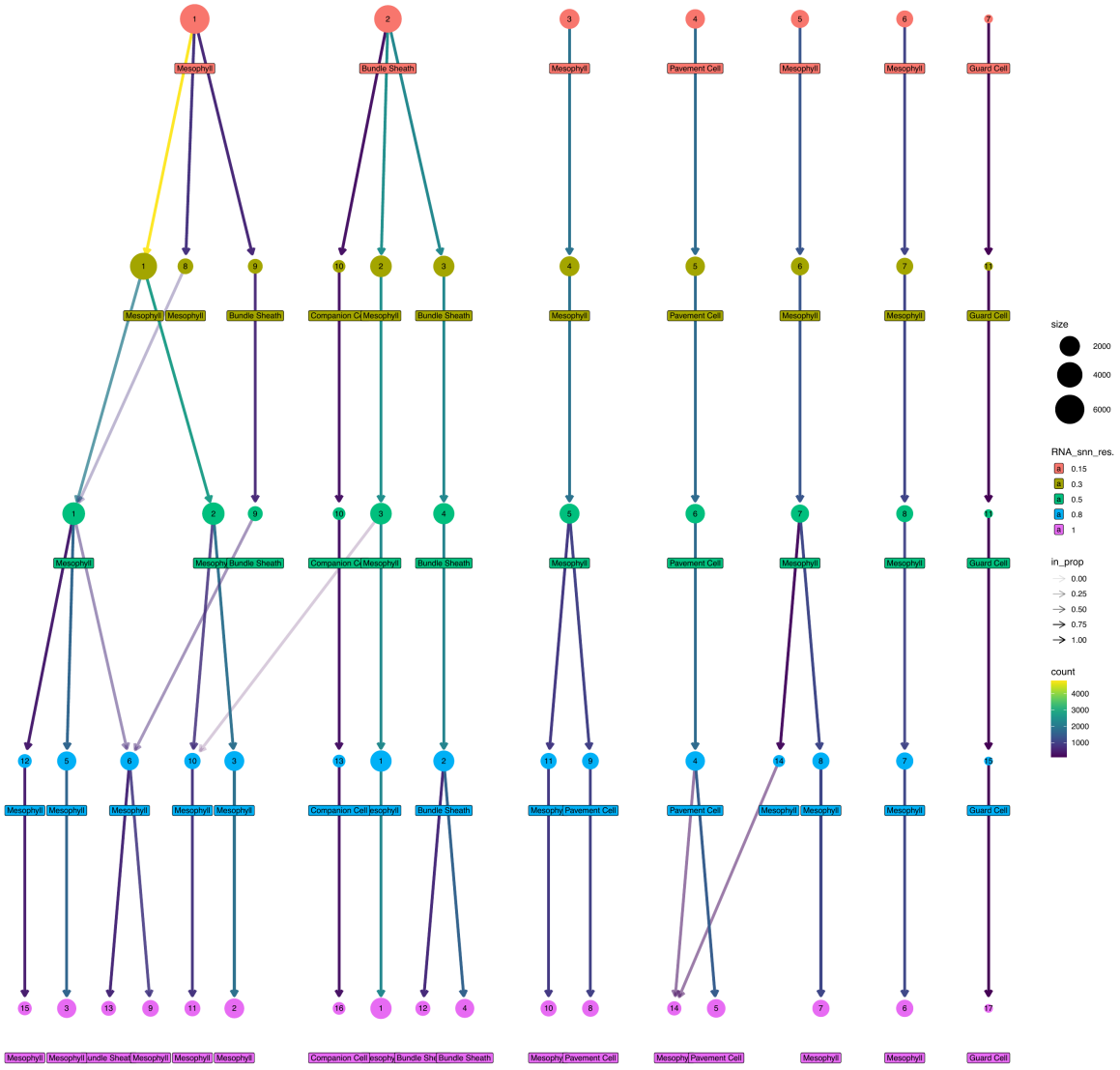


Figure 10: Cluster tree labeled by the dominant bibliography-based annotation in each node.

0.12 Section 9 - GENE EXPRESSION VISUALIZATION

```
# SECTION 9 - GENE EXPRESSION VISUALIZATION
# ============================================================================
```

```r
# Simple inspection for one gene: violin plot across all annotated cell types
# and UMAP expression overlay.
#
# +- SET GENE TO INSPECT ---------------------------------------------------
#   gene : single gene to inspect
# +-------------------------------------------------------------------------+
gene <- "AT5G26000"

output_dir <- dir_04

# JoinLayers is required in Seurat 5 after merging samples.
pbmc_harmony <- JoinLayers(pbmc_harmony)
Idents(pbmc_harmony) <- "celltype"

save_vln(
  VlnPlot(pbmc_harmony, features = gene),
  "vln_gene_all.pdf"
)

save_pdf(
  FeaturePlot(pbmc_harmony, features = gene),
  "feature_gene_all.pdf"
)

# ============================================================================

message("\nOK SECTION 9 COMPLETE: Gene expression visualization saved")
```

Representative outputs from Section 9:

0.13 Section 10 - CELL-TYPE GROUPING [OPTIONAL]

```r
# SECTION 10 - CELL-TYPE GROUPING  [OPTIONAL]
# ============================================================================
# Fine-grained labels are collapsed into broader categories for downstream
# analyses. Cell types NOT listed in 'grouping' keep their original label.
# Skip this section if you do not need to merge cell types.
#
# +- EDIT THIS MAP TO MATCH YOUR CELL TYPES --------------------------------
#   Left side  : original label (must match exactly)
#   Right side : new broader label to assign
# +-------------------------------------------------------------------------+
# Leave empty if you do not want to merge labels.
#
# Example: uncomment and edit this if Section 7 produced several labels that
# should be treated as one broader group downstream.
#
# grouping <- c(
#   "Epidermis Hypocotyl.1" = "Epidermis Hypocotyl",
#   "Epidermis Hypocotyl.2" = "Epidermis Hypocotyl",
#   "Epidermis Hypocotyl.3" = "Epidermis Hypocotyl",
#   "ENDO1" = "Endodermis",
#   "ENDO2" = "Endodermis",
```

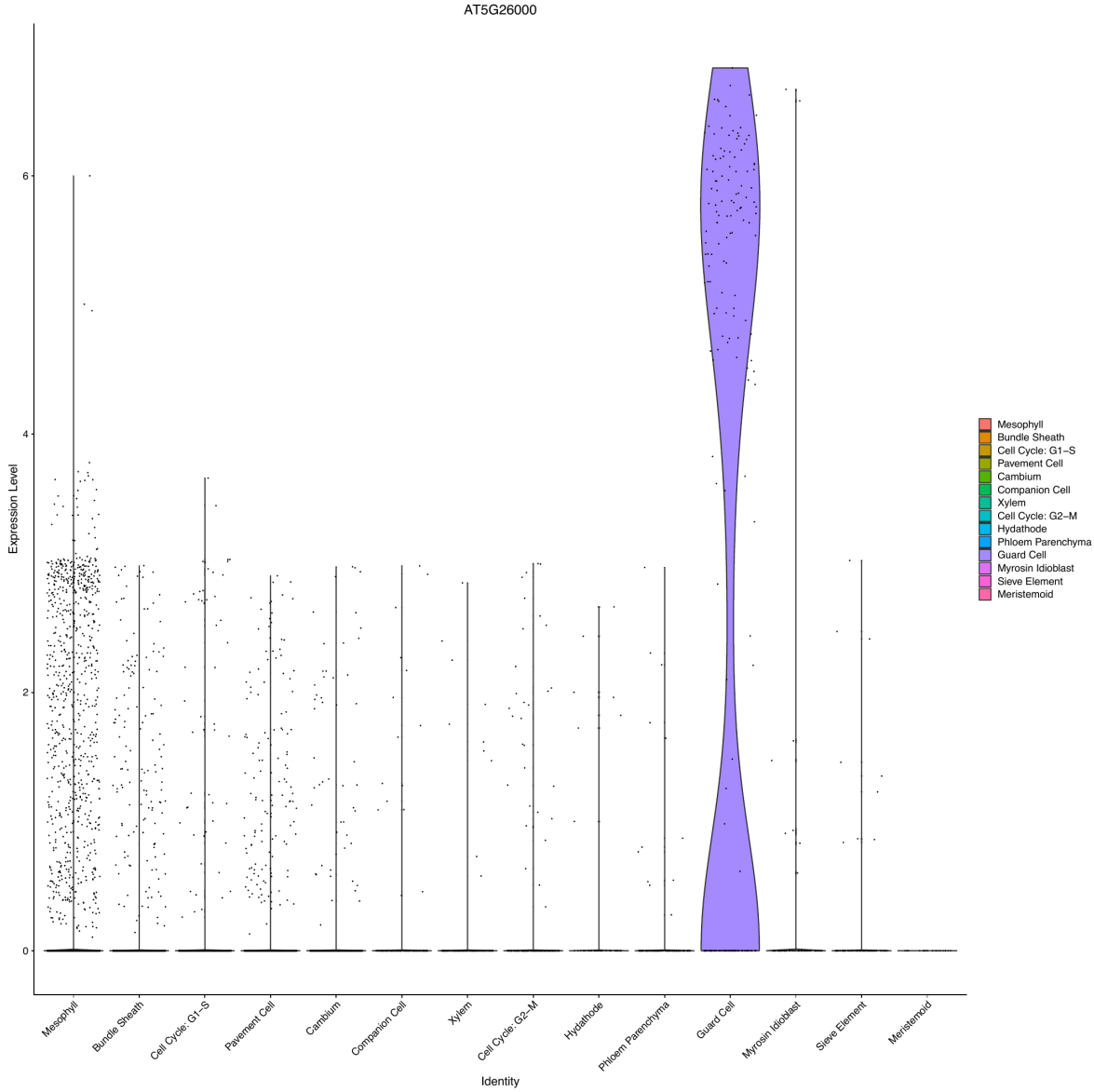


Figure 11: Violin plot for the selected gene across all annotated cell types.

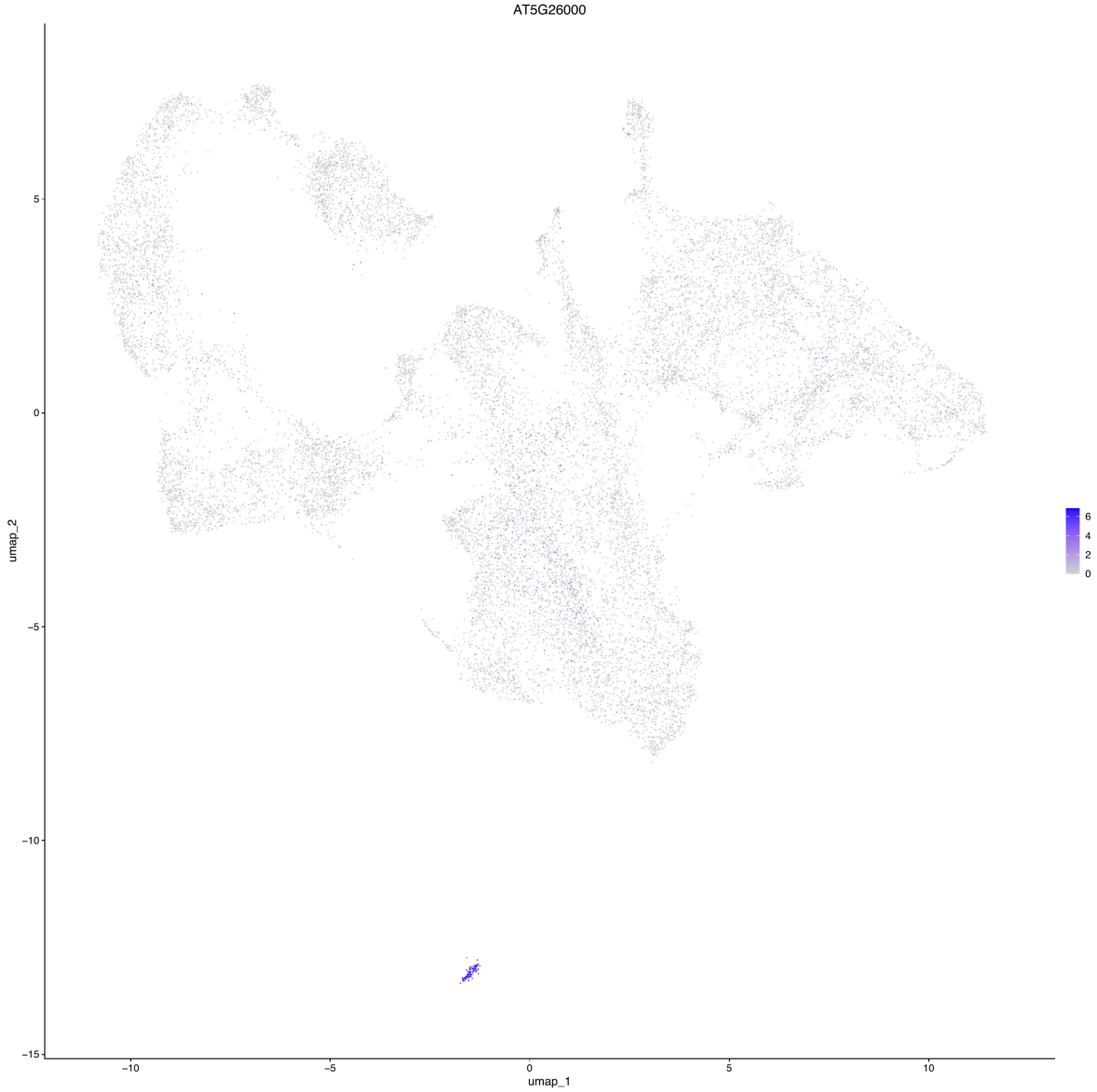


Figure 12: UMAP expression overlay for the selected gene.

```
#   "PHLOEM AND SIEVE ELEMENT" = "PHLOEM"
# )
#
# Current default: no grouping; celltype_grouped is a straight pass-through
# of celltype.
grouping <- c()

output_dir <- dir_05

# !!! unpacks the grouping vector as named arguments to recode()
# recode() errors on an empty replacement list, so an empty `grouping` (no
# merging needed) falls back to a straight pass-through instead.
if (length(grouping) > 0) {
  pbmc_harmony$celltype_grouped <- recode(pbmc_harmony$celltype, !!!grouping)
} else {
  pbmc_harmony$celltype_grouped <- pbmc_harmony$celltype
}

save_pdf(
  DimPlot(pbmc_harmony, group.by = "celltype_grouped",  # grouped broad cell types
          label = TRUE, repel = TRUE, raster = FALSE),
  "umap_grouped.pdf"
)


# ==============================================================================

message("\nOK SECTION 10 COMPLETE: Cell-type grouping complete")
```

Representative output from Section 10:

0.14 Section 11 - SUBCLUSTER MARKER INSPECTION [OPTIONAL]

This section is intentionally interactive. It is used to test marker expression against subclusters before deciding whether manual curation is needed. The section does not modify `pbmc_harmony` by default. Individual subcluster UMAPs are saved in a wide format so that UMAP dimensions keep the same visual proportions and do not appear compressed in the report.

```
# SECTION 11 - SUBCLUSTER MARKER INSPECTION  [OPTIONAL]
# ==============================================================================
# !!! WARNING: Run this section interactively, step by step.
# !!! Do NOT source the entire script with this section active.
# Skip this section if you are satisfied with the annotation from Section 7.
#
# Purpose: pick one or more annotated cell types, subcluster them, and inspect
# marker expression across the subclusters before deciding whether any manual
# reassignment is needed. This section does NOT modify pbmc_harmony by default.

output_dir   <- dir_05
curation_col <- "celltype"
Idents(pbmc_harmony) <- curation_col

print(table(pbmc_harmony[[curation_col]], useNA = "ifany"))
```

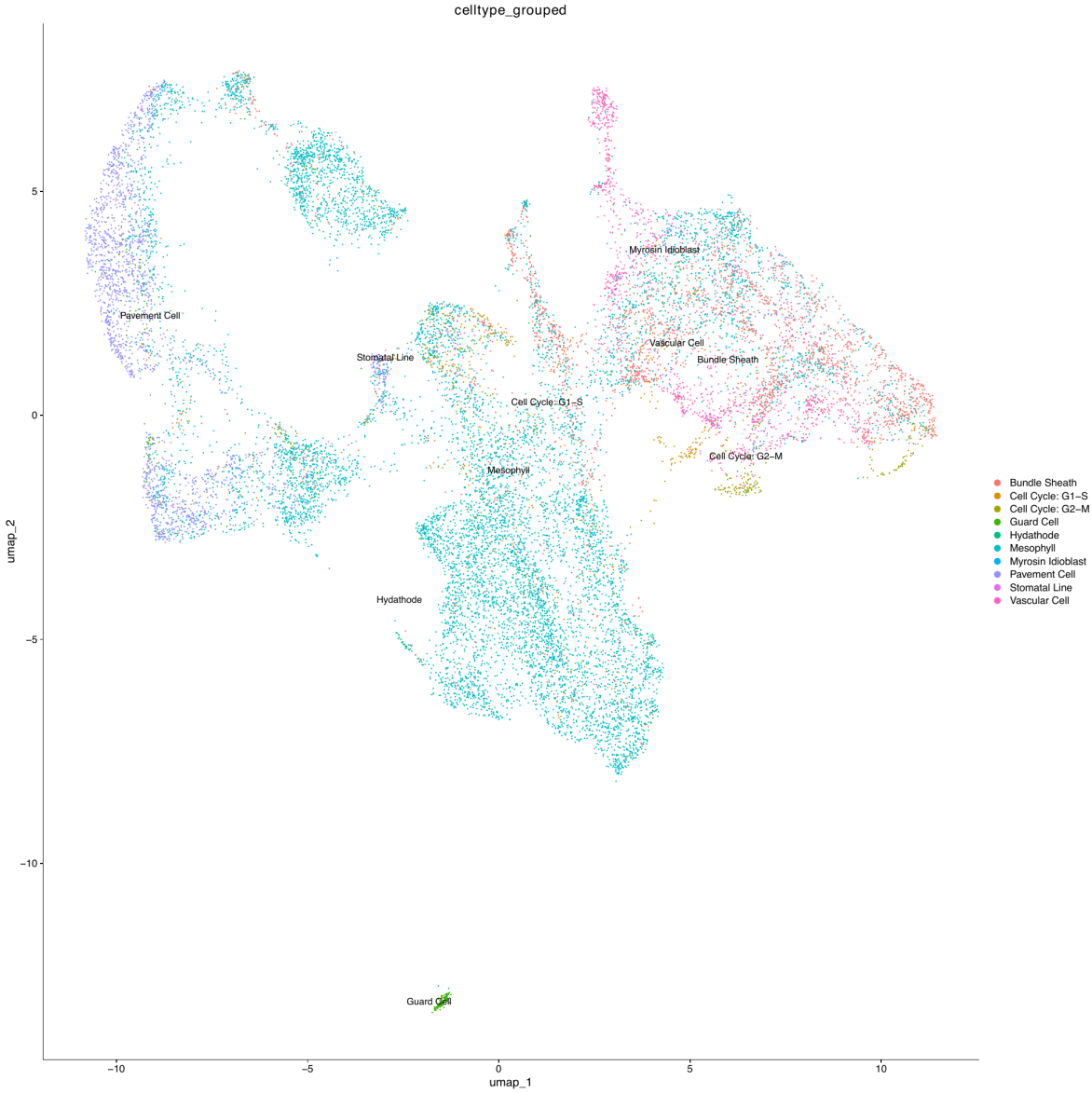


Figure 13: UMAP colored by grouped cell-type labels.

```r
# Edit this vector using exact names from the table above.
# Example options:
#   types_to_inspect <- c("Epidermis Hypocotyl.1")
#   types_to_inspect <- c("Epidermis Hypocotyl.1", "Epidermis Hypocotyl.2")
#   types_to_inspect <- names(table(pbmc_harmony[[curation_col]]))  # all labels; can be slow

types_to_inspect <- c("2","4","5","8")

subcluster_resolution <- 0.3
subcluster_dims       <- 1:20

subcluster_list    <- list()
inspection_entries <- list()

for (ct in types_to_inspect) {
  safe_name <- make.names(ct)
  file_tag  <- gsub("[^A-Za-z0-9]+", "_", ct)

  message("\nInspecting: ", ct)

  sub_obj <- subcluster_cell_type(
    pbmc_harmony,
    tipo       = ct,
    annot_col  = curation_col,
    resolution = subcluster_resolution,
    dims       = subcluster_dims
  )

  subcluster_list[[safe_name]] <- sub_obj
  print(table(sub_obj$cluster_subtipo, useNA = "ifany"))

  p_dim <- plot_subcluster_umap(sub_obj, ct, output_dir)

  plot_marker_dotplot(
    sub_obj,
    marker_table,
    annot_col = "cluster_subtipo",
    outfile   = file.path(output_dir, paste0("dotplot_markers_subclusters_", file_tag, ".pdf")),
    width = 18,
    height = 18
  )

  inspection_entries[[safe_name]] <- list(
    umap_plot = p_dim,
    obj       = sub_obj
  )
}

save_subcluster_composite(
  subcluster_list = inspection_entries,
  marker_table    = marker_table,
  output_dir      = output_dir,
  filename        = "subclustering_marker_inspection2.pdf"
```

```
)

# After reviewing the PDFs, manually decide whether to reassign subclusters.
# Nothing is changed in pbmc_harmony unless you run an apply_subcluster_reassignment()
# block intentionally.

# Example reassignment template, left commented on purpose:
#
# reassign <- list(
#   Epidermis.Hypocotyl.1 = c(
#     "0"      = "Epidermis Hypocotyl",
#     "1"      = "Epidermis Cotyledon",
#     "others" = "Epidermis Hypocotyl"
#   )
# )
#
# pbmc_harmony <- apply_subcluster_reassignment(
#   obj             = pbmc_harmony,
#   subcluster_list = subcluster_list,
#   reassign        = reassign,
#   source_col      = curation_col,
#   dest_col        = "celltype_curated"
# )
#
# save_pdf(
#   DimPlot(pbmc_harmony, group.by = "celltype_curated",
#           label = TRUE, repel = TRUE, raster = FALSE),
#   "umap_curated.pdf"
# )
# saveRDS(pbmc_harmony, file.path(dir_objects, "pbmc_harmony_curated.rds"))

# ==============================================================================

message("\nOK SECTION 11 COMPLETE: Subcluster marker inspection saved")
```

Representative output from Section 11:

0.15 Current stopping point

This methods draft stops at optional subcluster marker inspection. Export to h5ad, pseudobulk, DESeq2, enrichment, network analysis, and pseudotime are not included in this version.

0.16 Supplementary information

0.16.1 Minimum components to run on a personal computer

The minimum practical setup for reproducing Chapter 1 on a local workstation is:

Component	Minimum	Recommended
Operating system	Linux, macOS, or WSL2	Linux server or macOS with Docker Desktop
CPU cores	4	16 or more
RAM	32 GB	64-128 GB for merged multi-sample objects

Component	Minimum	Recommended
Disk space	100 GB free	500 GB or more when keeping FASTQ, Cell Ranger output, and R objects
Software	Docker	Docker plus external Cell Ranger installation
Inputs	Cell Ranger `filtered_feature_bc_matrix/` per sample	Raw FASTQ plus Cell Ranger output retained for provenance

Small pilot datasets can be run on a laptop or desktop if memory is sufficient. For full experiments, repeated clustering sweeps, Cell Ranger processing from FASTQ, or large merged Seurat objects, a server or external cluster is strongly recommended. Cell Ranger is especially CPU- and memory-intensive and is not installed inside the downstream `matigara/scrnaseq:latest` Docker image.

0.16.2 Docker image program and package versions

The following version tables were generated from the active Docker container used to render this methods draft. The complete machine-readable files are kept next to this R Markdown file:

```
reports/docker_cli_versions.tsv
reports/docker_r_package_versions.tsv
reports/docker_python_package_versions.tsv
```

Table 3: Command-line tools detected in the Docker image.

program	version
R	R version 4.5.3 (2026-03-11) – Reassured Reassurer
Rscript	Rscript (R) version 4.5.3 (2026-03-11)
python3	Python 3.12.3
pip	pip 26.1.2 from /opt/venv/lib/python3.12/site-packages/pip (python 3.12)
pandoc	pandoc 3.1.3
cellranger	not installed in Docker image
samtools	not installed in Docker image
bedtools	not installed in Docker image
bowtie2	not installed in Docker image
bwa	not installed in Docker image
STAR	not installed in Docker image
salmon	not installed in Docker image
kallisto	not installed in Docker image
fastqc	not installed in Docker image
multiqc	not installed in Docker image

Table 4: Complete R package inventory in the Docker image.

Package	Version	LibPath
abind	1.4-8	/usr/local/lib/R/site-library
anndataR	1.0.2	/usr/local/lib/R/site-library
AnnotationDbi	1.72.0	/usr/local/lib/R/site-library
ape	5.8-1	/usr/local/lib/R/site-library
aplot	0.2.9	/usr/local/lib/R/site-library

Package	Version	LibPath
askpass	1.2.1	/usr/local/lib/R/site-library
assertthat	0.2.1	/usr/local/lib/R/site-library
assorthead	1.4.0	/usr/local/lib/R/site-library
backports	1.5.1	/usr/local/lib/R/site-library
base	4.5.3	/usr/local/lib/R/library
base64enc	0.1-6	/usr/local/lib/R/site-library
basilisk	1.22.0	/usr/local/lib/R/site-library
batchelor	1.26.0	/usr/local/lib/R/site-library
beachmat	2.26.0	/usr/local/lib/R/site-library
beeswarm	0.4.0	/usr/local/lib/R/site-library
BH	1.90.0-1	/usr/local/lib/R/site-library
biglm	0.9-3	/usr/local/lib/R/site-library
Biobase	2.70.0	/usr/local/lib/R/site-library
BiocGenerics	0.56.0	/usr/local/lib/R/site-library
BiocManager	1.30.27	/usr/local/lib/R/site-library
BiocNeighbors	2.4.0	/usr/local/lib/R/site-library
BiocParallel	1.44.0	/usr/local/lib/R/site-library
BiocSingular	1.26.1	/usr/local/lib/R/site-library
BiocVersion	3.22.0	/usr/local/lib/R/site-library
Biostrings	2.78.0	/usr/local/lib/R/site-library
bit	4.6.0	/usr/local/lib/R/site-library
bit64	4.8.2	/usr/local/lib/R/site-library
bitops	1.0-9	/usr/local/lib/R/site-library
blob	1.3.0	/usr/local/lib/R/site-library
boot	1.3-32	/usr/local/lib/R/site-library
boot	1.3-32	/usr/local/lib/R/library
BPCells	0.3.1	/usr/local/lib/R/site-library
brew	1.0-10	/usr/local/lib/R/site-library
brio	1.1.5	/usr/local/lib/R/site-library
broom	1.0.13	/usr/local/lib/R/site-library
bslib	0.11.0	/usr/local/lib/R/site-library
cachem	1.1.0	/usr/local/lib/R/site-library
Cairo	1.7-0	/usr/local/lib/R/site-library
callr	3.7.6	/usr/local/lib/R/site-library
car	3.1-5	/usr/local/lib/R/site-library
carData	3.0-6	/usr/local/lib/R/site-library
caTools	1.18.3	/usr/local/lib/R/site-library
cellranger	1.1.0	/usr/local/lib/R/site-library
checkmate	2.3.4	/usr/local/lib/R/site-library
circlize	0.4.18	/usr/local/lib/R/site-library
class	7.3-23	/usr/local/lib/R/library
classInt	0.4-11	/usr/local/lib/R/site-library
cli	3.6.6	/usr/local/lib/R/site-library
clipr	0.8.1	/usr/local/lib/R/site-library
clue	0.3-68	/usr/local/lib/R/site-library
cluster	2.1.8.2	/usr/local/lib/R/library
clusterProfiler	4.18.4	/usr/local/lib/R/site-library
clustree	0.5.1	/usr/local/lib/R/site-library
codetools	0.2-20	/usr/local/lib/R/library
colorspace	2.1-2	/usr/local/lib/R/site-library
commonmark	2.0.0	/usr/local/lib/R/site-library
compiler	4.5.3	/usr/local/lib/R/library

Package	Version	LibPath
ComplexHeatmap	2.26.1	/usr/local/lib/R/site-library
conflicted	1.2.0	/usr/local/lib/R/site-library
corrplot	0.95	/usr/local/lib/R/site-library
cowplot	1.2.0	/usr/local/lib/R/site-library
cpp11	0.5.5	/usr/local/lib/R/site-library
crayon	1.5.3	/usr/local/lib/R/site-library
credentials	2.0.3	/usr/local/lib/R/site-library
crosstalk	1.2.2	/usr/local/lib/R/site-library
curl	7.1.0	/usr/local/lib/R/site-library
data.table	1.18.4	/usr/local/lib/R/site-library
datasets	4.5.3	/usr/local/lib/R/library
DBI	1.3.0	/usr/local/lib/R/site-library
dbplyr	2.5.2	/usr/local/lib/R/site-library
DelayedArray	0.36.1	/usr/local/lib/R/site-library
DelayedMatrixStats	1.32.0	/usr/local/lib/R/site-library
deldir	2.0-4	/usr/local/lib/R/site-library
Deriv	4.2.0	/usr/local/lib/R/site-library
desc	1.4.3	/usr/local/lib/R/site-library
DESeq2	1.50.2	/usr/local/lib/R/site-library
devtools	2.5.2	/usr/local/lib/R/site-library
diffobj	0.3.6	/usr/local/lib/R/site-library
digest	0.6.39	/usr/local/lib/R/site-library
dir.expiry	1.18.0	/usr/local/lib/R/site-library
distributional	0.7.0	/usr/local/lib/R/site-library
doBy	4.7.1	/usr/local/lib/R/site-library
docopt	0.7.2	/usr/local/lib/R/site-library
doParallel	1.0.17	/usr/local/lib/R/site-library
DOSE	4.4.0	/usr/local/lib/R/site-library
dotCall64	1.2	/usr/local/lib/R/site-library
DoubletFinder	2.0.6	/usr/local/lib/R/site-library
downlit	0.4.5	/usr/local/lib/R/site-library
dplyr	1.2.1	/usr/local/lib/R/site-library
dqrng	0.4.1	/usr/local/lib/R/site-library
dtplyr	1.3.3	/usr/local/lib/R/site-library
dynamicTreeCut	1.63-1	/usr/local/lib/R/site-library
e1071	1.7-17	/usr/local/lib/R/site-library
ellipsis	0.3.3	/usr/local/lib/R/site-library
enrichplot	1.30.5	/usr/local/lib/R/site-library
enrichR	3.4	/usr/local/lib/R/site-library
eulerr	8.0.0	/usr/local/lib/R/site-library
evaluate	1.0.5	/usr/local/lib/R/site-library
fansi	1.0.7	/usr/local/lib/R/site-library
farver	2.1.2	/usr/local/lib/R/site-library
fastcluster	1.3.0	/usr/local/lib/R/site-library
fastDummies	1.7.6	/usr/local/lib/R/site-library
fastmap	1.2.0	/usr/local/lib/R/site-library
fastmatch	1.1-8	/usr/local/lib/R/site-library
fgsea	1.36.2	/usr/local/lib/R/site-library
fields	17.3	/usr/local/lib/R/site-library
filelock	1.0.3	/usr/local/lib/R/site-library
fitdistrplus	1.2-6	/usr/local/lib/R/site-library
FNN	1.1.4.1	/usr/local/lib/R/site-library

Package	Version	LibPath
fontawesome	0.5.3	/usr/local/lib/R/site-library
fontBitstreamVera	0.1.1	/usr/local/lib/R/site-library
fontLiberation	0.1.0	/usr/local/lib/R/site-library
fontquiver	0.2.1	/usr/local/lib/R/site-library
forcats	1.0.1	/usr/local/lib/R/site-library
foreach	1.5.2	/usr/local/lib/R/site-library
forecast	9.0.2	/usr/local/lib/R/site-library
foreign	0.8-91	/usr/local/lib/R/library
formatR	1.14	/usr/local/lib/R/site-library
Formula	1.2-5	/usr/local/lib/R/site-library
fracdiff	1.5-4	/usr/local/lib/R/site-library
fs	2.1.0	/usr/local/lib/R/site-library
furrr	0.4.0	/usr/local/lib/R/site-library
futile.logger	1.4.9	/usr/local/lib/R/site-library
futile.options	1.0.1	/usr/local/lib/R/site-library
future	1.70.0	/usr/local/lib/R/site-library
future.apply	1.20.2	/usr/local/lib/R/site-library
gargle	1.6.1	/usr/local/lib/R/site-library
gdtools	0.5.1	/usr/local/lib/R/site-library
GeneOverlap	1.46.0	/usr/local/lib/R/site-library
generics	0.1.4	/usr/local/lib/R/site-library
GENIE3	1.32.0	/usr/local/lib/R/site-library
GenomeInfoDb	1.46.2	/usr/local/lib/R/site-library
GenomicRanges	1.62.1	/usr/local/lib/R/site-library
gert	2.3.1	/usr/local/lib/R/site-library
GetoptLong	1.1.1	/usr/local/lib/R/site-library
ggbeeswarm	0.7.3	/usr/local/lib/R/site-library
ggdist	3.3.3	/usr/local/lib/R/site-library
ggforce	0.5.0	/usr/local/lib/R/site-library
ggfun	0.2.0	/usr/local/lib/R/site-library
ggiraph	0.9.6	/usr/local/lib/R/site-library
ggnewscale	0.5.2	/usr/local/lib/R/site-library
ggplot2	4.0.3	/usr/local/lib/R/site-library
ggplotify	0.1.3	/usr/local/lib/R/site-library
ggpubr	0.6.3	/usr/local/lib/R/site-library
ggraph	2.2.2	/usr/local/lib/R/site-library
ggrastr	1.0.2	/usr/local/lib/R/site-library
ggrepel	0.9.8	/usr/local/lib/R/site-library
ggridges	0.5.7	/usr/local/lib/R/site-library
ggsci	5.0.0	/usr/local/lib/R/site-library
ggsignif	0.6.4	/usr/local/lib/R/site-library
ggtangle	0.1.2	/usr/local/lib/R/site-library
ggtree	4.0.5	/usr/local/lib/R/site-library
ggvenn	0.1.19	/usr/local/lib/R/site-library
gh	1.6.0	/usr/local/lib/R/site-library
gitcreds	0.1.2	/usr/local/lib/R/site-library
GlobalOptions	0.1.4	/usr/local/lib/R/site-library
globals	0.19.1	/usr/local/lib/R/site-library
glue	1.8.1	/usr/local/lib/R/site-library
GO.db	3.22.0	/usr/local/lib/R/site-library
goftest	1.2-3	/usr/local/lib/R/site-library
googledrive	2.1.2	/usr/local/lib/R/site-library

Package	Version	LibPath
googlesheets4	1.1.2	/usr/local/lib/R/site-library
GOSemSim	2.36.0	/usr/local/lib/R/site-library
gplots	3.3.0	/usr/local/lib/R/site-library
graphics	4.5.3	/usr/local/lib/R/library
graphlayouts	1.2.3	/usr/local/lib/R/site-library
grDevices	4.5.3	/usr/local/lib/R/library
grid	4.5.3	/usr/local/lib/R/library
gridExtra	2.3	/usr/local/lib/R/site-library
gridGraphics	0.5-1	/usr/local/lib/R/site-library
grr	0.9.5	/usr/local/lib/R/site-library
gson	0.1.0	/usr/local/lib/R/site-library
gtable	0.3.6	/usr/local/lib/R/site-library
gtools	3.9.5	/usr/local/lib/R/site-library
h5mread	1.2.1	/usr/local/lib/R/site-library
harmony	2.0.3	/usr/local/lib/R/site-library
haven	2.5.5	/usr/local/lib/R/site-library
HDF5Array	1.38.0	/usr/local/lib/R/site-library
hdf5r	1.3.12	/usr/local/lib/R/site-library
hdWGCNA	0.4.11	/usr/local/lib/R/site-library
here	1.0.2	/usr/local/lib/R/site-library
hexbin	1.28.5	/usr/local/lib/R/site-library
highr	0.12	/usr/local/lib/R/site-library
Hmisc	5.2-5	/usr/local/lib/R/site-library
hms	1.1.4	/usr/local/lib/R/site-library
htmlTable	2.5.0	/usr/local/lib/R/site-library
htmltools	0.5.9	/usr/local/lib/R/site-library
htmlwidgets	1.6.4	/usr/local/lib/R/site-library
httpuv	1.6.17	/usr/local/lib/R/site-library
httr	1.4.8	/usr/local/lib/R/site-library
httr2	1.2.2	/usr/local/lib/R/site-library
ica	1.0-3	/usr/local/lib/R/site-library
ids	1.0.1	/usr/local/lib/R/site-library
igraph	2.3.2	/usr/local/lib/R/site-library
impute	1.84.0	/usr/local/lib/R/site-library
ini	0.3.1	/usr/local/lib/R/site-library
IRanges	2.44.0	/usr/local/lib/R/site-library
irlba	2.3.7	/usr/local/lib/R/site-library
isoband	0.3.0	/usr/local/lib/R/site-library
iterators	1.0.14	/usr/local/lib/R/site-library
jquerylib	0.1.4	/usr/local/lib/R/site-library
jsonlite	2.0.0	/usr/local/lib/R/site-library
kableExtra	1.4.0	/usr/local/lib/R/site-library
KEGGREST	1.50.0	/usr/local/lib/R/site-library
KernSmooth	2.23-26	/usr/local/lib/R/site-library
KernSmooth	2.23-26	/usr/local/lib/R/library
knitr	1.51	/usr/local/lib/R/site-library
labeling	0.4.3	/usr/local/lib/R/site-library
lambda.r	1.2.4	/usr/local/lib/R/site-library
later	1.4.8	/usr/local/lib/R/site-library
lattice	0.22-9	/usr/local/lib/R/library
lazyeval	0.2.3	/usr/local/lib/R/site-library
leidenbase	0.1.37	/usr/local/lib/R/site-library

Package	Version	LibPath
lifecycle	1.0.5	/usr/local/lib/R/site-library
limma	3.66.0	/usr/local/lib/R/site-library
listenv	0.10.1	/usr/local/lib/R/site-library
littler	0.3.23	/usr/local/lib/R/site-library
lme4	2.0-1	/usr/local/lib/R/site-library
lmtest	0.9-40	/usr/local/lib/R/site-library
locfit	1.5-9.12	/usr/local/lib/R/site-library
lubridate	1.9.5	/usr/local/lib/R/site-library
magrittr	2.0.5	/usr/local/lib/R/site-library
maps	3.4.3	/usr/local/lib/R/site-library
MASS	7.3-65	/usr/local/lib/R/site-library
MASS	7.3-65	/usr/local/lib/R/library
Matrix	1.7-5	/usr/local/lib/R/site-library
Matrix	1.7-4	/usr/local/lib/R/library
MatrixGenerics	1.22.0	/usr/local/lib/R/site-library
MatrixModels	0.5-4	/usr/local/lib/R/site-library
matrixStats	1.5.0	/usr/local/lib/R/site-library
memoise	2.0.1	/usr/local/lib/R/site-library
methods	4.5.3	/usr/local/lib/R/library
mgcv	1.9-4	/usr/local/lib/R/library
microbenchmark	1.5.0	/usr/local/lib/R/site-library
mime	0.13	/usr/local/lib/R/site-library
miniUI	0.1.2	/usr/local/lib/R/site-library
minqa	1.2.8	/usr/local/lib/R/site-library
modelr	0.1.11	/usr/local/lib/R/site-library
monocle3	1.4.27	/usr/local/lib/R/site-library
nlme	3.1-169	/usr/local/lib/R/site-library
nlme	3.1-168	/usr/local/lib/R/library
nloptr	2.2.1	/usr/local/lib/R/site-library
nnet	7.3-20	/usr/local/lib/R/library
numDeriv	2016.8-1.1	/usr/local/lib/R/site-library
openssl	2.4.1	/usr/local/lib/R/site-library
org.At.tair.db	3.22.0	/usr/local/lib/R/site-library
otel	0.2.0	/usr/local/lib/R/site-library
pak	0.9.5	/usr/local/lib/R/site-library
parallel	4.5.3	/usr/local/lib/R/library
parallelly	1.47.0	/usr/local/lib/R/site-library
patchwork	1.3.2	/usr/local/lib/R/site-library
pbapply	1.7-4	/usr/local/lib/R/site-library
pbkrtest	0.5.5	/usr/local/lib/R/site-library
pbmcapply	1.5.1	/usr/local/lib/R/site-library
pheatmap	1.0.13	/usr/local/lib/R/site-library
pillar	1.11.1	/usr/local/lib/R/site-library
pkgbuild	1.4.8	/usr/local/lib/R/site-library
pkgconfig	2.0.3	/usr/local/lib/R/site-library
pkgdown	2.2.0	/usr/local/lib/R/site-library
pkgload	1.5.2	/usr/local/lib/R/site-library
plotly	4.12.0	/usr/local/lib/R/site-library
plyr	1.8.9	/usr/local/lib/R/site-library
png	0.1-9	/usr/local/lib/R/site-library
polyclip	1.10-7	/usr/local/lib/R/site-library
polynom	1.4-1	/usr/local/lib/R/site-library

Package	Version	LibPath
praise	1.0.0	/usr/local/lib/R/site-library
preprocessCore	1.72.0	/usr/local/lib/R/site-library
prettyunits	1.2.0	/usr/local/lib/R/site-library
processx	3.9.0	/usr/local/lib/R/site-library
profvis	0.4.0	/usr/local/lib/R/site-library
progress	1.2.3	/usr/local/lib/R/site-library
progressr	0.19.0	/usr/local/lib/R/site-library
promises	1.5.0	/usr/local/lib/R/site-library
proxy	0.4-29	/usr/local/lib/R/site-library
ps	1.9.3	/usr/local/lib/R/site-library
pscl	1.5.9	/usr/local/lib/R/site-library
purrr	1.2.2	/usr/local/lib/R/site-library
quadprog	1.5-8	/usr/local/lib/R/site-library
quantreg	6.1	/usr/local/lib/R/site-library
qvalue	2.42.0	/usr/local/lib/R/site-library
R.methodsS3	1.8.2	/usr/local/lib/R/site-library
R.oo	1.27.1	/usr/local/lib/R/site-library
R.utils	2.13.0	/usr/local/lib/R/site-library
R6	2.6.1	/usr/local/lib/R/site-library
ragg	1.5.2	/usr/local/lib/R/site-library
RANN	2.6.2	/usr/local/lib/R/site-library
rappdirs	0.3.4	/usr/local/lib/R/site-library
rbibutils	2.4.1	/usr/local/lib/R/site-library
rcmdcheck	1.4.0	/usr/local/lib/R/site-library
RColorBrewer	1.1-3	/usr/local/lib/R/site-library
Rcpp	1.1.1-1.1	/usr/local/lib/R/site-library
RcppAnnoy	0.0.23	/usr/local/lib/R/site-library
RcppArmadillo	15.2.7-1	/usr/local/lib/R/site-library
RcppEigen	0.3.4.0.2	/usr/local/lib/R/site-library
RcppHNSW	0.7.0	/usr/local/lib/R/site-library
RcppML	0.3.7.1	/usr/local/lib/R/site-library
RcppProgress	0.4.2	/usr/local/lib/R/site-library
RcppRoll	0.3.2	/usr/local/lib/R/site-library
RcppTOML	0.2.3	/usr/local/lib/R/site-library
Rdpack	2.6.6	/usr/local/lib/R/site-library
readr	2.2.0	/usr/local/lib/R/site-library
readxl	1.5.0	/usr/local/lib/R/site-library
reformulas	0.4.4	/usr/local/lib/R/site-library
rematch	2.0.0	/usr/local/lib/R/site-library
rematch2	2.1.2	/usr/local/lib/R/site-library
remotes	2.5.0	/usr/local/lib/R/site-library
reprex	2.1.1	/usr/local/lib/R/site-library
reshape2	1.4.5	/usr/local/lib/R/site-library
ResidualMatrix	1.20.0	/usr/local/lib/R/site-library
reticulate	1.46.0	/usr/local/lib/R/site-library
rhdf5	2.54.1	/usr/local/lib/R/site-library
rhdf5filters	1.22.0	/usr/local/lib/R/site-library
Rhdf5lib	1.32.0	/usr/local/lib/R/site-library
RhpcBLASctl	0.23-42	/usr/local/lib/R/site-library
Rhtslib	3.6.0	/usr/local/lib/R/site-library
rjson	0.2.23	/usr/local/lib/R/site-library
rlang	1.2.0	/usr/local/lib/R/site-library

Package	Version	LibPath
rmarkdown	2.31	/usr/local/lib/R/site-library
ROCR	1.0-12	/usr/local/lib/R/site-library
roxygen2	8.0.0	/usr/local/lib/R/site-library
rpart	4.1.24	/usr/local/lib/R/library
rprojroot	2.1.1	/usr/local/lib/R/site-library
rsample	1.3.2	/usr/local/lib/R/site-library
Rsamtools	2.26.0	/usr/local/lib/R/site-library
RSpectra	0.16-2	/usr/local/lib/R/site-library
RSQLite	3.53.1	/usr/local/lib/R/site-library
rstatix	0.7.3	/usr/local/lib/R/site-library
rstudioapi	0.18.0	/usr/local/lib/R/site-library
rsvd	1.0.5	/usr/local/lib/R/site-library
Rtsne	0.17	/usr/local/lib/R/site-library
rversions	3.0.0	/usr/local/lib/R/site-library
rvest	1.0.5	/usr/local/lib/R/site-library
s2	1.1.11	/usr/local/lib/R/site-library
S4Arrays	1.10.1	/usr/local/lib/R/site-library
S4Vectors	0.48.1	/usr/local/lib/R/site-library
S7	0.2.2	/usr/local/lib/R/site-library
sass	0.4.10	/usr/local/lib/R/site-library
ScaledMatrix	1.18.0	/usr/local/lib/R/site-library
scales	1.4.0	/usr/local/lib/R/site-library
scater	1.38.1	/usr/local/lib/R/site-library
scattermore	1.2	/usr/local/lib/R/site-library
scatterpie	0.2.6	/usr/local/lib/R/site-library
sctransform	0.4.3	/usr/local/lib/R/site-library
scuttle	1.20.0	/usr/local/lib/R/site-library
selectr	0.5-1	/usr/local/lib/R/site-library
Seqinfo	1.0.0	/usr/local/lib/R/site-library
sessioninfo	1.2.3	/usr/local/lib/R/site-library
Seurat	5.5.0	/usr/local/lib/R/site-library
SeuratDisk	0.0.0.9021	/usr/local/lib/R/site-library
SeuratObject	5.4.0	/usr/local/lib/R/site-library
SeuratWrappers	0.4.0	/usr/local/lib/R/site-library
sf	1.1-1	/usr/local/lib/R/site-library
shape	1.4.6.1	/usr/local/lib/R/site-library
shiny	1.13.0	/usr/local/lib/R/site-library
SingleCellExperiment	1.32.0	/usr/local/lib/R/site-library
sitmo	2.0.2	/usr/local/lib/R/site-library
slam	0.1-55	/usr/local/lib/R/site-library
slider	0.3.3	/usr/local/lib/R/site-library
snow	0.4-4	/usr/local/lib/R/site-library
sourcetools	0.1.7-2	/usr/local/lib/R/site-library
sp	2.2-1	/usr/local/lib/R/site-library
spam	2.11-4	/usr/local/lib/R/site-library
SparseArray	1.10.10	/usr/local/lib/R/site-library
SparseM	1.84-2	/usr/local/lib/R/site-library
sparseMatrixStats	1.22.0	/usr/local/lib/R/site-library
spatial	7.3-18	/usr/local/lib/R/library
spatstat.data	3.1-9	/usr/local/lib/R/site-library
spatstat.explore	3.8-1	/usr/local/lib/R/site-library
spatstat.geom	3.8-1	/usr/local/lib/R/site-library

Package	Version	LibPath
spatstat.random	3.5-0	/usr/local/lib/R/site-library
spatstat.sparse	3.2-0	/usr/local/lib/R/site-library
spatstat.univar	3.2-0	/usr/local/lib/R/site-library
spatstat.utils	3.2-3	/usr/local/lib/R/site-library
spData	2.3.5	/usr/local/lib/R/site-library
spdep	1.4-2	/usr/local/lib/R/site-library
speedglm	0.3-4	/usr/local/lib/R/site-library
splines	4.5.3	/usr/local/lib/R/library
statmod	1.5.2	/usr/local/lib/R/site-library
stats	4.5.3	/usr/local/lib/R/library
stats4	4.5.3	/usr/local/lib/R/library
stringi	1.8.7	/usr/local/lib/R/site-library
stringr	1.6.0	/usr/local/lib/R/site-library
SummarizedExperiment	1.40.0	/usr/local/lib/R/site-library
survival	3.8-6	/usr/local/lib/R/site-library
survival	3.8-6	/usr/local/lib/R/library
svglite	2.2.2	/usr/local/lib/R/site-library
sys	3.4.3	/usr/local/lib/R/site-library
systemfonts	1.3.2	/usr/local/lib/R/site-library
tcltk	4.5.3	/usr/local/lib/R/library
tensor	1.5.1	/usr/local/lib/R/site-library
tester	0.3.0	/usr/local/lib/R/site-library
testthat	3.3.2	/usr/local/lib/R/site-library
textshaping	1.0.5	/usr/local/lib/R/site-library
tibble	3.3.1	/usr/local/lib/R/site-library
tidydr	0.0.6	/usr/local/lib/R/site-library
tidygraph	1.3.1	/usr/local/lib/R/site-library
tidyr	1.3.2	/usr/local/lib/R/site-library
tidyselect	1.2.1	/usr/local/lib/R/site-library
tidytree	0.4.7	/usr/local/lib/R/site-library
tidyverse	2.0.0	/usr/local/lib/R/site-library
timechange	0.4.0	/usr/local/lib/R/site-library
timeDate	4052.112	/usr/local/lib/R/site-library
tinytex	0.59	/usr/local/lib/R/site-library
tools	4.5.3	/usr/local/lib/R/library
treeio	1.34.0	/usr/local/lib/R/site-library
tweenr	2.0.3	/usr/local/lib/R/site-library
tzdb	0.5.0	/usr/local/lib/R/site-library
UCell	2.14.0	/usr/local/lib/R/site-library
UCSC.utils	1.6.1	/usr/local/lib/R/site-library
units	1.0-1	/usr/local/lib/R/site-library
UpSetR	1.4.1	/usr/local/lib/R/site-library
urca	1.3-4	/usr/local/lib/R/site-library
urlchecker	1.0.1	/usr/local/lib/R/site-library
usethis	3.2.1	/usr/local/lib/R/site-library
utf8	1.2.6	/usr/local/lib/R/site-library
utils	4.5.3	/usr/local/lib/R/library
uuid	1.2-2	/usr/local/lib/R/site-library
uwot	0.2.4	/usr/local/lib/R/site-library
vctrs	0.7.3	/usr/local/lib/R/site-library
VennDiagram	1.8.2	/usr/local/lib/R/site-library
vipor	0.4.7	/usr/local/lib/R/site-library

Package	Version	LibPath
viridis	0.6.5	/usr/local/lib/R/site-library
viridisLite	0.4.3	/usr/local/lib/R/site-library
vroom	1.7.1	/usr/local/lib/R/site-library
waldo	0.6.2	/usr/local/lib/R/site-library
warp	0.2.3	/usr/local/lib/R/site-library
WGCNA	1.74	/usr/local/lib/R/site-library
whisker	0.4.1	/usr/local/lib/R/site-library
withr	3.0.2	/usr/local/lib/R/site-library
wk	0.9.5	/usr/local/lib/R/site-library
WriteXLS	6.8.0	/usr/local/lib/R/site-library
xfun	0.58	/usr/local/lib/R/site-library
xml2	1.5.2	/usr/local/lib/R/site-library
xopen	1.0.1	/usr/local/lib/R/site-library
xtable	1.8-8	/usr/local/lib/R/site-library
XVector	0.50.0	/usr/local/lib/R/site-library
yaml	2.3.12	/usr/local/lib/R/site-library
yulab.utils	0.2.4	/usr/local/lib/R/site-library
zellkonverter	1.20.1	/usr/local/lib/R/site-library
zip	2.3.3	/usr/local/lib/R/site-library
zoo	1.8-15	/usr/local/lib/R/site-library

Table 5: Complete Python package inventory in the Docker image.

Package	Version
adjustText	1.3.0
anndata	0.12.16
annotated-types	0.7.0
anyio	4.13.0
argon2-cffi	25.1.0
argon2-cffi-bindings	25.1.0
array-api-compat	1.14.0
arrow	1.4.0
asttokens	3.0.1
async-lru	2.3.0
attrs	26.1.0
babel	2.18.0
beautifulsoup4	4.14.3
bleach	6.3.0
blosc2	4.4.1
cellbender	0.3.2
certifi	2026.5.20
cffi	2.0.0
charset-normalizer	3.4.7
click	8.4.1
comm	0.2.3
contourpy	1.3.3
cuda-bindings	13.3.1
cuda-pathfinder	1.5.5
cuda-toolkit	13.0.2
cycler	0.12.1

Package	Version
debugpy	1.8.21
decorator	5.3.1
defusedxml	0.7.1
donfig	0.8.1.post1
elpigraph-python	0.3.4
entrypoints	0.4
executing	2.2.1
fa2_modified	0.4
fast-array-utils	1.4.1
fastjsonschema	2.21.2
filelock	3.29.0
fonttools	4.63.0
fqdn	1.5.1
fsspec	2026.4.0
google-crc32c	1.8.0
h11	0.16.0
h5py	3.16.0
httpcore	1.0.9
httpx	0.28.1
idna	3.18
igraph	1.0.0
ipykernel	6.29.5
ipython	9.14.0
ipython-genutils	0.2.0
ipython_pygments_lexers	1.1.1
ipywidgets	8.1.8
isoduration	20.11.0
jax	0.10.1
jaxlib	0.10.1
jaxopt	0.8.5
jedi	0.20.0
Jinja2	3.1.6
joblib	1.5.3
json5	0.14.0
jsonpointer	3.1.1
jsonschema	4.26.0
jsonschema-specifications	2025.9.1
jupyter	1.1.1
jupyter-console	6.6.3
jupyter-events	0.12.1
jupyter-highlight-selected-word	0.2.0
jupyter-lsp	2.3.1
jupyter_client	7.4.9
jupyter_contrib_core	0.4.2
jupyter_contrib_nbextensions	0.7.0
jupyter_core	5.9.1
jupyter_nbextensions_configurator	0.6.4
jupyter_server	2.19.0
jupyter_server_terminals	0.5.4
jupyterlab	4.5.7
jupyterlab_pygments	0.3.0
jupyterlab_server	2.28.0

Package	Version
jupyterlab_widgets	3.0.16
kiwisolver	1.5.0
lark	1.3.1
legacy-api-wrap	1.5
leidenalg	0.12.0
llvmlite	0.47.0
loompy	3.0.8
lxml	6.1.1
lxml_html_clean	0.4.5
MarkupSafe	3.0.3
matplotlib	3.10.9
matplotlib-inline	0.2.2
mellon	1.7.1
mistune	0.8.4
ml_dtypes	0.5.4
mpmath	1.3.0
msgpack	1.1.2
narwhals	2.22.0
natsort	8.4.0
nbclassic	1.3.3
nbclient	0.10.4
nbconvert	6.5.4
nbformat	5.10.4
ndindex	1.10.1
nest-asyncio	1.6.0
networkx	3.6.1
notebook	6.5.7
notebook_shim	0.2.4
numba	0.65.1
numcodecs	0.16.5
numexpr	2.14.1
numpy	2.4.6
numpy-groupies	0.11.3
nvidia-cublas	13.1.1.3
nvidia-cuda-cupti	13.0.85
nvidia-cuda-nvrtc	13.0.88
nvidia-cuda-runtime	13.0.96
nvidia-cudnn-cu13	9.20.0.48
nvidia-cufft	12.0.0.61
nvidia-cufile	1.15.1.6
nvidia-curand	10.4.0.35
nvidia-cusolver	12.0.4.66
nvidia-cusparse	12.6.3.3
nvidia-cusparselt-cu13	0.8.1
nvidia-nccl-cu13	2.29.7
nvidia-nvjitlink	13.0.88
nvidia-nvshmem-cu13	3.4.5
nvidia-nvtx	13.0.85
opt_einsum	3.4.0
packaging	26.2
palantir	1.4.4
pandas	2.3.3

Package	Version
pandocfilters	1.5.1
parso	0.8.7
patsy	1.0.2
pexpect	4.9.0
pillow	12.2.0
pip	26.1.2
platformdirs	4.10.0
plotly	6.7.0
prometheus_client	0.25.0
prompt_toolkit	3.0.52
psutil	7.2.2
ptyprocess	0.7.0
pure_eval	0.2.3
py-cpuinfo	9.0.0
pycparser	3.0
pydantic	2.13.4
pydantic_core	2.46.4
Pygments	2.20.0
pynndescent	0.6.0
pyparsing	3.3.2
pyro-api	0.1.2
pyro-ppl	1.9.1
python-dateutil	2.9.0.post0
python-igraph	1.0.0
python-json-logger	4.1.0
pytz	2026.2
PyYAML	6.0.3
pyzmq	27.1.0
referencing	0.37.0
requests	2.34.2
rfc3339-validator	0.1.4
rfc3986-validator	0.1.1
rfc3987-syntax	1.1.0
rpds-py	2026.5.1
rpy2	3.6.7
rpy2-rinterface	3.6.6
rpy2-robjects	3.6.5
scanpy	1.12.1
scFates	1.2.4
scikit-learn	1.9.0
scikit-misc	0.5.2
scipy	1.17.1
scverse-misc	0.0.7
seaborn	0.13.2
Send2Trash	2.1.0
session-info2	0.4.1
setuptools	81.0.0
shapely	2.1.2
simpleppt	1.1.7
six	1.17.0
soupsieve	2.8.4
stack-data	0.6.3

Package	Version
statsmodels	0.14.6
sympy	1.14.0
tables	3.11.1
terminado	0.18.1
texttable	1.7.0
threadpoolctl	3.6.0
tinycss2	1.5.1
torch	2.12.0
tornado	6.5.6
tqdm	4.67.3
traitlets	5.15.0
triton	3.7.0
typing-inspection	0.4.2
typing_extensions	4.15.0
tzdata	2026.2
tzlocal	5.4.3
umap-learn	0.5.12
uri-template	1.3.0
urllib3	2.7.0
wcwidth	0.7.0
webcolors	25.10.0
webencodings	0.5.1
websocket-client	1.9.0
widgetsnbextension	4.0.15
zarr	3.2.1

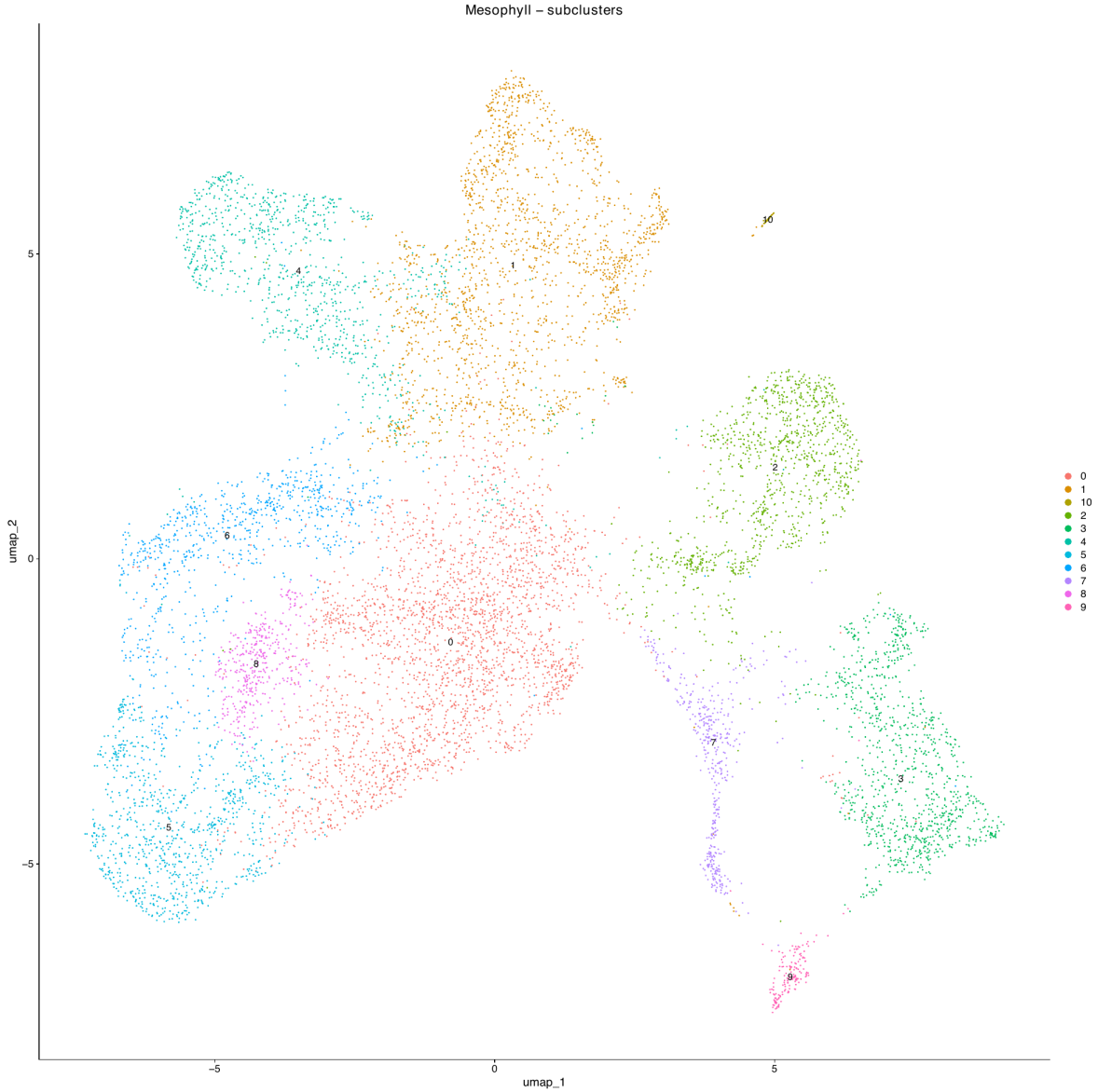


Figure 14: Example subcluster UMAP used for manual marker inspection.